

Actuarial Non-Life Insurance
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These slides have been written
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- **Prerequisites.** The prerequisites for these lectures are a solid background in mathematics (***in particular, in probability theory and statistics***), and to have attended (and studied) the basic course in *Actuarial Mathematics (only Non-Life module)/Introduction to Actuarial Mathematics and Risk Theory (non-life part)*.
- **Terms of Use.** These slides are an ongoing project which will continuously be revised and updated. Of course, there may be typos (or errors) in the notes and there is always room for improvement. Therefore, we appreciate any comment and/or corrections that readers may have.

Teaching

- Lectures integrated with examples from actuarial case studies.
- Excel files
- Seminars/Business Game

- **Exam**

The exam is based on **both** a written and an oral examination.

1. The written examination is based on both open-ended questions and exercises. It could be with the use of Excel or calculator. Further information will be given during the course
 2. The compulsory oral exam can be taken only if the written exam has been passed (i.e. with a mark greater or equal than 18).
- In case the exam has been failed, both written and oral examination must be taken again.
 - The examination regards the entire course syllabus.
 - The examination is aimed at assessing reasoning and analytical rigour with regard to course topics.
 - *Students, that have not passed the exam of basic Actuarial Mathematics or Introduction to Actuarial Mathematics and Risk Theory, cannot attend the exam (only non-life module is requested)*

Agenda (1/2)

- **1. Introduction: the general framework**
- **2. Deterministic Methods**
 - **2.1 Notation and Development Triangles**
 - **2.2 Paid Chain-Ladder (brief recall)**
 - **2.3 Incurred Chain-Ladder**
 - **2.4 Expected Loss Ratio**
 - **2.5 (Paid and Incurred) Bornhuetter-Ferguson method**
 - **2.6 Fisher-Lange method (brief recall)**
 - **2.7 Frequency-Severity methods**
 - **2.8 Summary of results**

Agenda (2/2)

- **3. Market Consistent Valuation of technical liabilities (Solvency II Framework)**
 - 3.1 Best Estimate
 - 3.2 Discounting Rates
 - 3.3 Risk Margin
- **4. Stochastic Model for Claims Reserve**
 - 4.1 Prediction Error Estimation: Mack Model
 - 4.2 The Over-Dispersed Poisson Model
 - 4.3 Bootstrap for Claims Reserving
 - 4.4 Collective Risk Model
 - 4.5 One Year View: main approaches
 - 4.6 One Year Approach and Capital Requirement
 - 4.7 Case Study
- **5. Ratemaking in MTPL line of business**
- **A focus on Reinsurance Treaties (dott. Luini)**

Main References and Further Material (Optional)

- PART II
 - Casualty Actuarial Society, *Estimating Unpaid Claims using basic techniques*, 2009.
 - Olivieri, A., Pitacco, E., *Introduction to Insurance Mathematics, Technical and Financial Features of Risk Transfers*, Springer
- Part III
 - European Commission, *Commission Delegated Regulation (EU) 2015/35 of 10 October 2014 supplementing Directive 2009/138/EC of the European Parliament and of the Council on the taking-up and pursuit of the business of Insurance and Reinsurance (Solvency II) Text with EEA relevance*
 - Solvency II Directive (Directive 2009/138 EC and Directive 2014/51/EU)
- Part IV
 - P. England, R. Verrall, *Stochastic claims reserving in general insurance*, British Actuarial Journal 8/3, 443-518, 2002.
 - P. England, *Addendum to “Analytic and bootstrap estimates of prediction errors in claims reserving”*, Insurance Mathematics and Economics 31, 2002.
 - European Commission, *Commission Delegated Regulation (EU) 2015/35 of 10 October 2014 supplementing Directive 2009/138/EC of the European Parliament and of the Council on the taking-up and pursuit of the business of Insurance and Reinsurance (Solvency II) Text with EEA relevance*
 - T. MACK, *Distribution-free calculation of the standard error of chain ladder reserves estimates*, ASTIN Bulletin 23/2, 213-225, 1993.
 - M. MERZ, M. WÜTHRICH, *Modelling the claims development result for solvency purposes*, CAS E-Forum, Fall 2008, 542-568, 2008.
 - E. Ohlsson, J. Lauzeningsks, *The One-Year Non-Life Insurance Risk*
 - G. TAYLOR, *Loss Reserving: An Actuarial Perspective*, Kluwer, 2000, Boston.
 - G. TAYLOR, G. MCGUIRE, *Stochastic Loss Reserving using Generalized Linear Models*, CAS Monographs.
 - M.V. WÜTHRICH, M. MERZ, *Stochastic Claims Reserving Methods in Insurance*, Wiley Finance, 2008.
- Part V
 - S. A. Klugman, H. H. Panjer, G. E. Willmot, *Loss Models: From Data to Decisions*, 4th Edition, Wiley,
 - G. Werner, C. Modlin, *Basic Ratemaking*, CAS, 2016

1. Introduction

The General Framework

- We can define the Non-Life *Underwriting Profits and Losses* (PL_t) of year t for a specific line of business (LoB) as:

$$PL_t = B_t - X'_t - V_t^S + V_{t-1}^S - E_t = \\ (B'_t - V_t^P + V_{t-1}^P) - X'^{CY}_t - V_t^{S,CY} - E_t \\ + V_{t-1}^S - X'^{PY}_t - V_t^{S,PY}$$

Notice that: Reinsurance, financial profits and losses, aggregation between LoBs are not considered in this formula

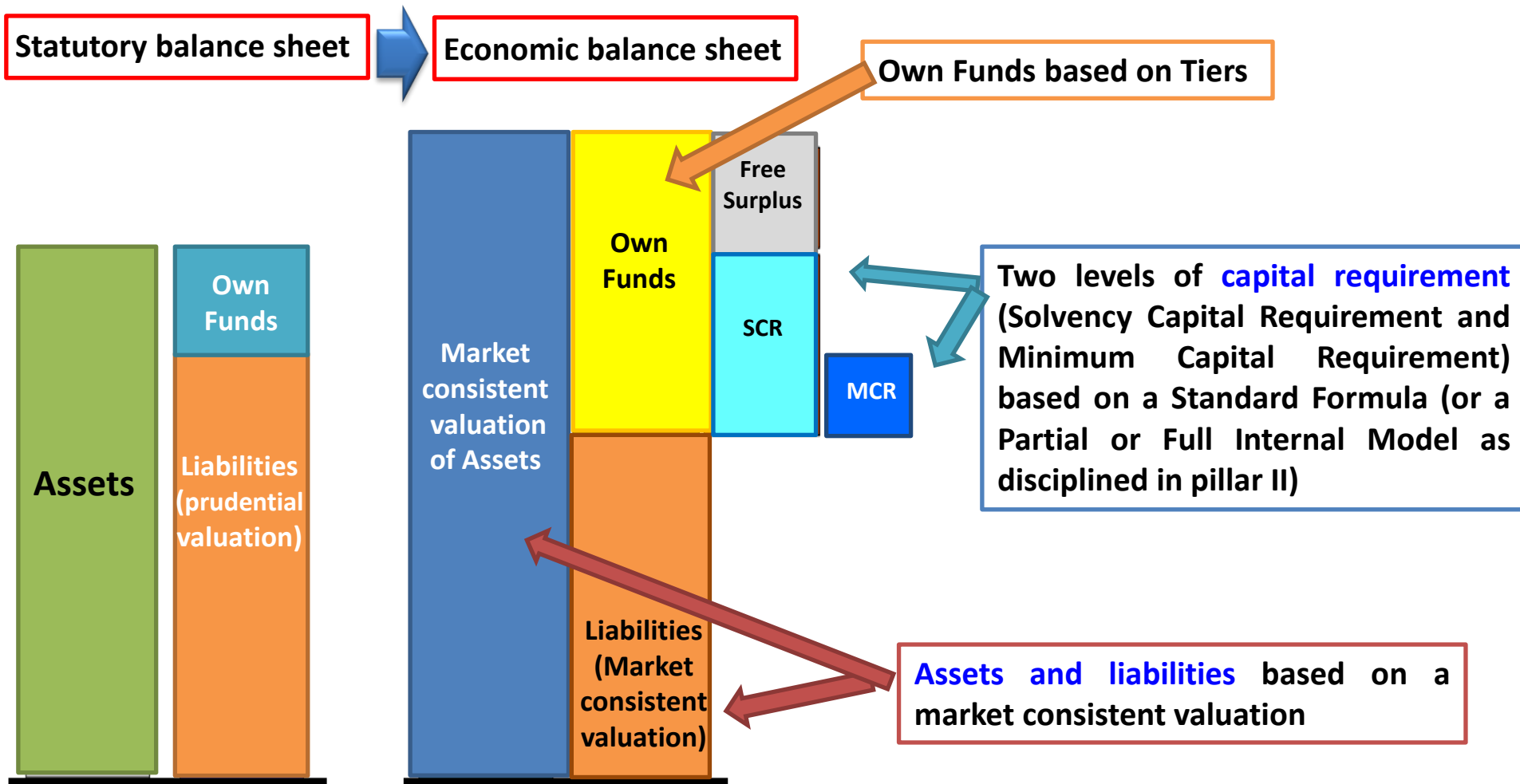
- Two alternative views: Ex-ante or ex-post valuation.
- To aim at:
 - Splitting expected value into two separate components (i.e. it allows to measure premium and reserve profit and loss).
 - Analysing distribution of different random variables involved (for instance useful for capital requirement assessment for NL underwriting risk)

Notice that X'_t represents payments for claims, while B_t represents earned premiums

Local Accounting vs Solvency II vs IFRS17

- Important remarks:
- Different valuation criteria have been provided in different balance sheets (local, IFRS or solvency report) for outstanding claims reserve and for premium reserve:
 - prudential vs market consistent valuation
- We need to take into account the evolution of solvency regulation (RSM vs ASM, SCR vs OF, etc.) in order to catch and explain the different effects on solvency ratios.

From statutory to an economic balance sheet



Local accounting criteria for technical reserves

- Premium Reserve = $UPR + URR$ (or $AURR$)
 - Unearned premium reserve (e.g. pro-rata)
 - Additional unexpired risk reserve (eventual extra amount to cover future obligations)
- Outstanding Claims Reserve
 - Not discounted
 - Gross of recoveries
 - Case outstanding (or unpaid case)
 - Provision for future development on known (reported) claims (RBNS)
 - Claim adjustment expenses (or claim related expenses): $ALAE + ULAE$
 - Provision for claims incurred but not reported (IBNR)
 - Adjustments (changing claims environment)

2.Deterministic Methods

2.1. Notation and Development Triangles

What kind of data?

- In general, reserve could be based on the following information:
 - *claims specific factors* like number of claims, severity of claim, business lines and policies involved, recovery process, payments made, claims adjusters' estimates, medical expertise, individual expert judgment, etc.
 - *external factors* like weather conditions, natural catastrophes (storm, flood, earthquake, etc.) state of economy, (claims) inflation, state of job market, potential legal changes, technological progress, political factors, global expert judgment, etc.;

Development Triangle and Notation

		Development Year (j)															
		0	1	2	3	...							n-1	n	n+1	n+2	...
Accident Year (i)	0	$X_{0,0}$	$X_{0,1}$														
	1	$X_{1,0}$															
	2																
	3																
	...																
n-1																	
n																	

observations (D_n)

$X_{i,j}$

to be predicted

Accident or reporting year (AY or RY):

$$i \in [0, n]$$

Development Year (DY):

$$j \in [0, n]$$

Calendar Year:

$$i + j$$

Upper Triangle:

$$D^{(n)} = \{X_{i,j} : i + j \leq n\}$$

Alternative Structures:

- Number of columns greater than number of rows (tail)
- Number of rows greater than number of columns (trapezoid)

Accounting (calendar) year k:

$$X_k = \sum_{i+j=k} X_{i,j} = \sum_{i=0}^k X_{i,k-i}$$

$X_{i,j}$ can represent amounts or numbers

Cumulative payments

- Incremental paid claims $P_{i,j}$  $C_{i,j} = \sum_{h=0}^j P_{i,h}$
- Incremental amount at Reserve (Case Outstanding or Booked Reserve, Best Estimate) $R_{i,j}$
- Number of reported claims $d_{i,j}$  d_i (aggregation by Report Year)
- Number of paid claims $n_{i,j}$
- Number of open claims $r_{i,j}$
- Reopened claim counts $re_{i,j}$
- Number of closed claims without payments (CNP counts) $s_{i,j}$

An Example SIFA Insurance

Earned Premiums B_t

1993	105,432
1994	108,635
1995	112,879
1996	118,932
1997	120,654
1998	141,755
1999	157,865
2000	158,323
2001	161,137
2002	165,782
2003	187,659
2004	194,567

Number of reported claims d_i

1993	57,133
1994	57,896
1995	58,721
1996	58,248
1997	57,785
1998	59,753
1999	58,772
2000	54,761
2001	49,832
2002	47,899
2003	49,511
2004	48,946

Amounts in thousand of Euro

Data of the example are organized by report year.

An Example SIFA Insurance

Incremental Paid Claims

 $P_{i,j}$

Amounts in thousand of Euro (.000)

Report Year (RY)	Development Year (DY)											
	0	1	2	3	4	5	6	7	8	9	10	11
1993	28,446	29,251	12,464	5,144	2,727	2,359	1,334	1,238	941	860	282	727
1994	31,963	36,106	13,441	5,868	2,882	2,422	918	1,076	734	458	456	
1995	37,775	40,125	12,951	6,034	3,010	1,264	1,250	1,135	904	559		
1996	40,418	44,499	15,370	5,594	2,616	1,984	2,137	1,184	873			
1997	44,116	45,490	15,339	5,478	2,541	2,906	1,294	1,124				
1998	50,294	48,040	17,843	7,035	3,934	2,726	2,267					
1999	49,620	49,991	19,570	10,047	5,750	3,313						
2000	46,410	49,694	20,881	8,202	4,714							
2001	48,295	49,354	18,304	8,833								
2002	52,590	50,606	18,604									
2003	58,599	53,743										
2004	60,361											

 $n_{i,j}$

RY / DY	0	1	2	3	4	5	6	7	8	9	10	11
1993	34,433	13,796	1,589	568	278	152	104	55	30	31	18	19
1994	35,475	13,718	1,501	548	209	134	51	44	27	20	16	
1995	37,004	13,820	1,527	436	194	72	46	29	17	17		
1996	37,038	13,631	1,463	500	164	80	63	40	26			
1997	36,849	13,416	1,564	422	182	107	81	42				
1998	39,171	12,601	1,592	559	273	175	155					
1999	37,492	12,282	2,057	739	391	287						
2000	34,188	12,245	1,938	761	295							
2001	31,308	10,743	1,908	639								
2002	30,357	10,117	1,611									
2003	30,717	11,081										
2004	30,590											

$$m_{i,j} = \frac{P_{i,j}}{n_{i,j}} \quad \text{Average paid cost (or average paid cost on claim counts with payment)}$$

$R_{i,j}$

Amounts in thousand of Euro

RY / DY	0	1	2	3	4	5	6	7	8	9	10	11
1993	56,551	26,907	12,910	7,836	6,201	4,555	3,806	3,189	2,792	1,987	1,784	1,068
1994	63,344	25,706	12,497	7,797	6,088	4,289	3,743	2,915	2,252	1,762	1,795	
1995	66,954	25,502	13,034	8,177	5,767	4,623	3,620	2,547	1,804	1,853		
1996	72,768	26,613	13,663	9,695	7,363	6,052	4,111	3,060	1,901			
1997	76,338	29,399	16,012	10,904	8,774	6,308	5,255	2,845				
1998	83,493	37,080	19,807	14,191	11,144	8,161	4,926					
1999	92,136	44,671	25,963	17,438	12,034	7,718						
2000	97,059	47,808	26,658	17,590	12,339							
2001	94,658	44,919	26,905	18,725								
2002	104,836	48,968	29,487									
2003	112,265	47,921										
2004	107,822											

 $r_{i,j}$

RY/DY	0	1	2	3	4	5	6	7	8	9	10	11
1993	19,508	4,038	1,374	727	445	289	180	125	98	71	54	36
1994	18,794	3,310	1,114	529	318	185	148	105	80	58	43	
1995	18,099	2,854	838	395	196	139	97	73	69	51		
1996	17,565	2,732	875	395	250	178	123	89	66			
1997	17,207	2,642	869	474	318	216	141	102				
1998	16,775	3,170	1,310	758	476	296	145					
1999	17,413	4,396	1,900	1,056	615	325						
2000	17,714	4,325	1,885	1,006	685							
2001	15,417	3,872	1,807	1,050								
2002	14,126	3,751	1,895									
2003	15,481	3,931										
2004	15,178											

$$r_{i,0} = d_i - n_{i,0} - s_{i,0}$$

$$r_{i,j} = r_{i,j-1} - n_{i,j} - s_{i,j} + re_{i,j}$$

In case of aggregation by accident year, relations need to also take into account of claims reported in DY greater than 0

$$S_{i,j}$$

RY / DY	0	1	2	3	4	5	6	7	8	9	10	11
1993	3,192	2,239	1,349	183	52	29	25	16	7	4	7	4
1994	3,627	2,306	1,002	151	56	33	11	14	7	7	7	
1995	3,617	2,034	846	124	61	19	12	8	4	6		
1996	3,645	1,875	754	126	38	25	15	11	7			
1997	3,729	1,858	679	129	45	29	24	13				
1998	3,806	2,136	768	148	76	60	21					
1999	3,866	1,934	988	290	142	55						
2000	2,859	2,137	1,044	303	113							
2001	3,107	1,952	824	299								
2002	3,416	1,664	874									
2003	3,313	1,763										
2004	3,178											

$$re_{i,j}$$

RY / DY	0	1	2	3	4	5	6	7	8	9	10	11
1993	-	565	275	104	48	25	19	16	9	8	8	5
1994	-	540	308	114	54	34	24	14	9	5	7	
1995	-	610	357	118	56	33	16	13	17	5		
1996	-	676	353	149	60	31	25	17	10			
1997	-	701	469	152	68	32	31	16				
1998	-	1,132	487	152	67	57	26					
1999	-	1,204	547	186	98	52						
2000	-	1,006	528	185	87							
2001	-	1,135	660	181								
2002	-	1,407	629									
2003	-	1,295										
2004	-											

The Development Triangle as a Diagnostic Tool

- Settlement speed (based on amounts or claim counts)
- Average Costs
- Ratio of Closed-to-Reported Claim Counts
- Ratio of CNP less Reopened Claims – to- Reported Claim Counts
- Ratio of closed claims with payments to total closed claims
- Run-off of claims reserve

2.2. Paid Chain-Ladder Method (A Brief Recall)

Background

- It is the most popular deterministic method for estimating outstanding claims.
- From the practical point of view, there are two arguments in favour of the chain ladder method: the method is simple and it exploits all data from the run-off triangle.
- The CL algorithm takes the position that the observations $D^{(n)}$ are extrapolated into the lower triangle.
- The quality of the chain ladder method depends on the stochastic mechanism, or stochastic model, generating the data.

Assumptions

- The use of run-off triangles for estimating future claims loss payments requires the assumption that the development of settled claims losses follows the same pattern for every claims occurrence year. Furthermore, it is assumed that this pattern will hold for future payments, i.e. the pattern will continue into the future.
- It also seems reasonable to expect that the estimates for settlement amounts in the future will be more accurate if all of the available data is used in the estimation. The Chain-Ladder Method is a technique that follows this expectation. It relies on the use of the cumulative claims loss settlement data for all claims occurrence years rather than considering, for example, only the latest claims occurrence year
- Main assumption:
 - All accident years i behave similarly and for cumulative claims we have approximatively
$$C_{i,j+1} \approx \lambda_j C_{i,j} \quad \forall i, j$$
for given factors $\lambda_j > 0$. These factors are called Chain-Ladder factors, average age-to-age factors, claim (or loss) development factors.

- The assumption of identical settlement patterns *should not be made lightly*.
- A basic requirement for using past experience to estimate future developments is that the data is accurate and free of errors. An insurer should also ensure that its own experience does not contradict the assumptions underlying the estimation method.
- Reasons why the observed claims loss settlement patterns may change over time include:
 - Changes in product design and conditions;
 - Changes in the claims reporting, assessment and settlement processes;
 - Changes in inflation rates;
 - Changes in the legal environment;
 - Abnormally large or small claim settlement amounts.
- If it is found that the past experience contradicts the assumptions of the method to be used then either the data must be appropriately adjusted or an alternative method must be used.

Incremental and Cumulative Payments

Incremental Paid Claims $P_{i,j}$

Amounts in thousand of Euro (.000)

Report Year (RY)	Development Year (DY)											
	0	1	2	3	4	5	6	7	8	9	10	11
1993	28,446	29,251	12,464	5,144	2,727	2,359	1,334	1,238	941	860	282	727
1994	31,963	36,106	13,441	5,868	2,882	2,422	918	1,076	734	458	456	
1995	37,775	40,125	12,951	6,034	3,010	1,264	1,250	1,135	904	559		
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1997	44,116	45,490	15,339	5,478	2,541	2,906	1,294	1,124				
1998	50,294	48,040	17,843	7,035	3,934	2,726	2,267					
1999	49,620	49,991	19,570	10,047	5,750	3,313						
2000	46,410	49,694	20,881	8,202	4,714							
2001	48,295	49,354	18,304	8,833								
2002	52,590	50,606	18,604									
2003	58,599	53,743										
2004	60,361											

Cumulative Paid Claims $C_{i,j}$

Amounts in thousand of Euro (.000)

	0	1	2	3	4	5	6	7	8	9	10	11	12+
1993	28,446	57,697	70,161	75,306	78,032	80,392	81,726	82,964	83,906	84,766	85,047	85,775	86,843
1994	31,963	68,069	81,511	87,379	90,261	92,683	93,601	94,677	95,411	95,869	96,325		
1995	37,775	77,900	90,851	96,885	99,895	101,159	102,409	103,544	104,448	105,007			
1996	40,418	84,917	100,287	105,881	108,496	110,481	112,617	113,802	114,674				
1997	44,116	89,606	104,946	110,424	112,964	115,871	117,165	118,289					
1998	50,294	98,335	116,177	123,213	127,146	129,872	132,139						
1999	49,620	99,611	119,181	129,228	134,978	138,291							
2000	46,410	96,104	116,985	125,186	129,900								
2001	48,295	97,648	115,953	124,786									
2002	52,590	103,196	121,800										
2003	58,599	112,342											
2004	60,361												

Development of Cumulative Claims

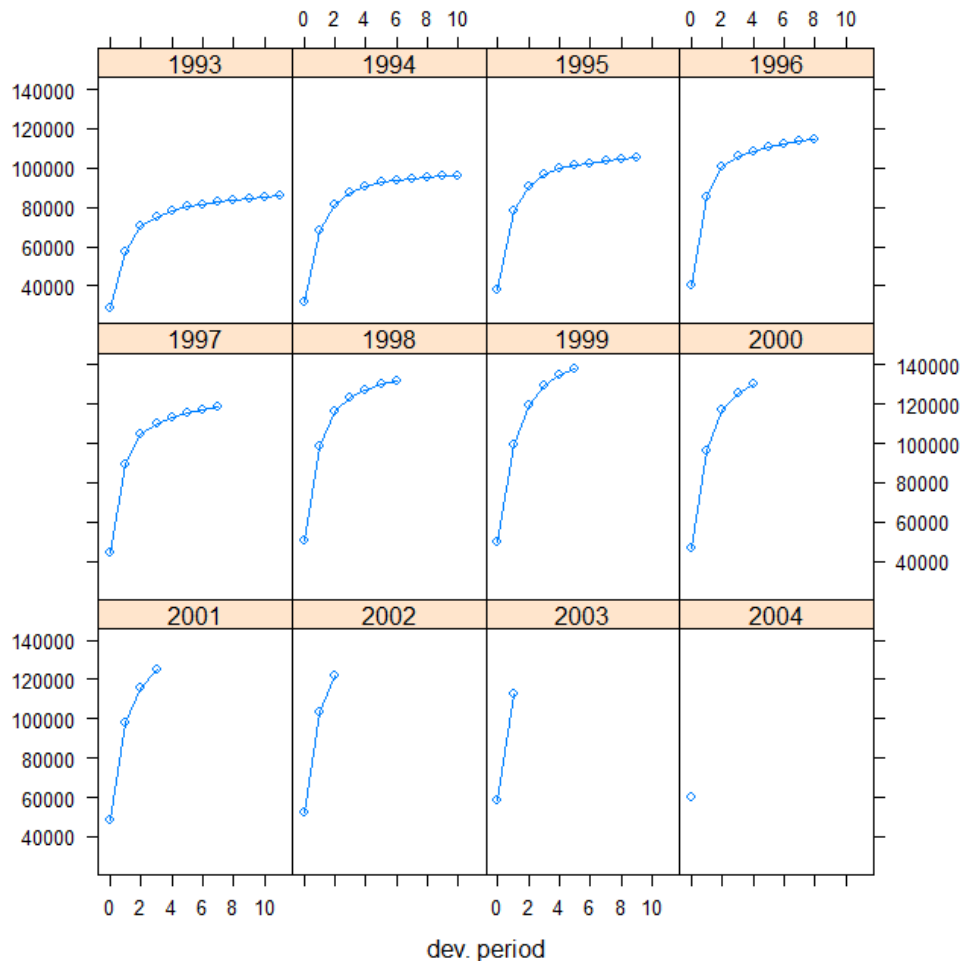


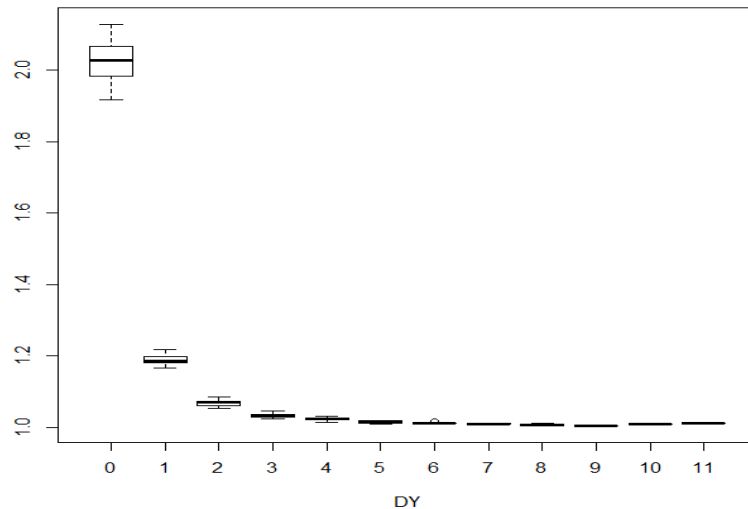
Figure illustrates the development of cumulative claims $C_{i,j} \in D^{(n)}$ for each i at time n .

From these graphs we immediately see that the data is rather regular but volume is increasing in accident years

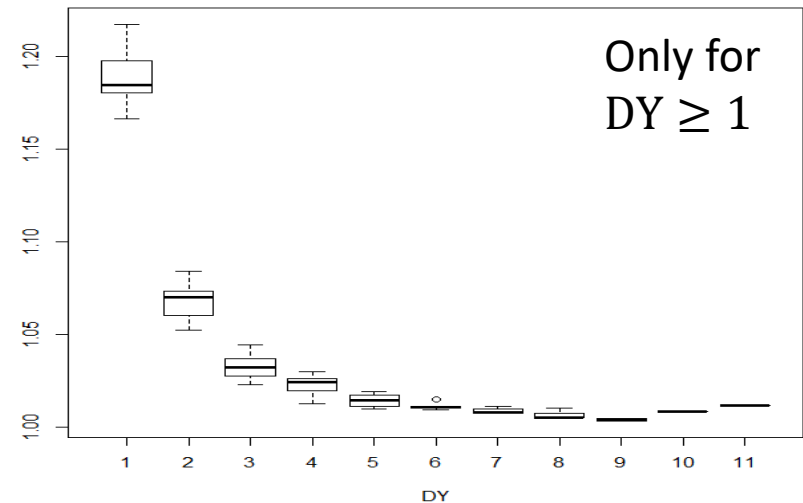
Individual loss development factors (age-to-age)

0	1	2	3	4	5	6	7	8	9	10	11
2.028	1.216	1.073	1.036	1.030	1.017	1.015	1.011	1.010	1.003	1.009	1.012
2.130	1.197	1.072	1.033	1.027	1.010	1.011	1.008	1.005	1.005		
2.062	1.166	1.066	1.031	1.013	1.012	1.011	1.009	1.005			
2.101	1.181	1.056	1.025	1.018	1.019	1.011	1.008				
2.031	1.171	1.052	1.023	1.026	1.011	1.010					
1.955	1.181	1.061	1.032	1.021	1.017						
2.007	1.196	1.084	1.044	1.025							
2.071	1.217	1.070	1.038								
2.022	1.187	1.076									
1.962	1.180										
1.917											

Individual Loss Development Factors



Individual Loss Development Factors



Development factors

The main difficulty is that, in general, the CL factors λ_j are not known and, henceforth, need to be estimated. Assuming that a volume weighted estimate provides the most reliable results we have:

$$\hat{\lambda}_j = \frac{\sum_{i=0}^{n-j-1} C_{i,j+1}}{\sum_{i=0}^{n-j-1} C_{i,j}} = \frac{\sum_{i=0}^{n-j-1} f_{i,j} C_{i,j}}{\sum_{i=0}^{n-j-1} C_{i,j}}, \text{ for } j = 0, \dots, n-1$$

Hence, we have a row vector $\hat{\lambda} = [\hat{\lambda}_0, \dots, \hat{\lambda}_{n-1}]$.

Alternative estimators (as simple average or the average of only last 3,4,5 diagonals) are often used in practice.

When selecting claim development factors, actuaries review the claim development experience for the following characteristics:

- Smooth progression of individual age-to-age factors and development factors across DYs;
- Stability of age-to-age factors for the same DY;
- Credibility of the experience;
- Applicability of historical experience;
- Changes in pattern.

Tail factor

- Furthermore, **tail factors** are used by actuaries to estimate the additional development that will occur after the eldest maturity in a given loss development triangle, or after the eldest credible link ratio (for details see for instance *The Estimation of Loss Development Tail Factors: A Summary Report*, CAS).
- Typical alternatives are:
 - Estimation of ultimate cost of first AY based on loss reserve
 - Curve fitting methods
 - Benchmark based methods
 - Other alternatives (Algebraic methods, bondy-type methods, etc).
- All methods allow to estimate $\hat{\lambda}_j$ for $j > n - 1$. Being Paid Chain-Ladder based on a multiplicative structure, all these estimates can be subsumed in a factor $\hat{\lambda}_\infty$. We have then $\hat{\lambda} = [\hat{\lambda}_0, \dots, \hat{\lambda}_{n-1}, \hat{\lambda}_\infty]$ when a tail factor is present.

Individual loss development vs development factors

0	1	2	3	4	5	6	7	8	9	10	11
2.028	1.216	1.073	1.036	1.030	1.017	1.015	1.011	1.010	1.003	1.009	1.012
2.130	1.197	1.072	1.033	1.027	1.010	1.011	1.008	1.005	1.005		
2.062	1.166	1.066	1.031	1.013	1.012	1.011	1.009	1.005			
2.101	1.181	1.056	1.025	1.018	1.019	1.011	1.008				
2.031	1.171	1.052	1.023	1.026	1.011	1.010					
1.955	1.181	1.061	1.032	1.021	1.017						
2.007	1.196	1.084	1.044	1.025							
2.071	1.217	1.070	1.038								
2.022	1.187	1.076									
1.962	1.180										
1.917											

	0	1	2	3	4	5	6	7	8	9	10	11
wtd	2.017	1.189	1.068	1.033	1.023	1.015	1.011	1.009	1.007	1.004	1.009	1.012
Simple av.	2.026	1.189	1.068	1.033	1.023	1.014	1.012	1.009	1.007	1.004	1.009	1.012
Last 4	1.993	1.195	1.073	1.034	1.022	1.015	1.011	1.009				

Lower Triangle and CL reserves

- Equipped with these CL factor estimates $\hat{\lambda}$, we predict cumulative payments i, j of lower triangle as:

$$\hat{C}_{i,j} = C_{i,n-i} \prod_{h=n-i}^{j-1} \hat{\lambda}_h$$

for $i = 1, \dots, n; j = n - i + 1, \dots, n$

- $\hat{C}_{i,n}$ represents the estimate of ultimate cost of AY i (if tail is not present). We have instead $\hat{C}_{i,\infty} = \hat{C}_{i,n} \hat{\lambda}_{\infty}, \forall i$ when a tail is added
- The undiscounted claim reserve, derived by Paid Chain-Ladder method, is:

$$R^{PCL} = \sum_{i=0}^n (\hat{C}_{i,\infty} - C_{i,n-i})$$

where $\hat{C}_{i,\infty} = \hat{C}_{i,n}$ if tail is not present

Chain-Ladder Predicted Cumulative Payments

$C_{i,j}$



	0	1	2	3	4	5	6	7	8	9	10	11	12+
1993	28,446	57,697	70,161	75,306	78,032	80,392	81,726	82,964	83,906	84,766	85,047	85,775	86,843
1994	31,963	68,069	81,511	87,379	90,261	92,683	93,601	94,677	95,411	95,869	96,325	97,149	98,358
1995	37,775	77,900	90,851	96,885	99,895	101,159	102,409	103,544	104,448	105,007	105,437	106,338	107,662
1996	40,418	84,917	100,287	105,881	108,496	110,481	112,617	113,802	114,674	115,433	115,905	116,896	118,351
1997	44,116	89,606	104,946	110,424	112,964	115,871	117,165	118,289	119,323	120,113	120,604	121,635	123,149
1998	50,294	98,335	116,177	123,213	127,146	129,872	132,139	133,638	134,806	135,698	136,252	137,417	139,128
1999	49,620	99,611	119,181	129,228	134,978	138,291	140,309	141,901	143,141	144,088	144,677	145,914	147,731
2000	46,410	96,104	116,985	125,186	129,900	132,834	134,772	136,301	137,492	138,402	138,967	140,156	141,901
2001	48,295	97,648	115,953	124,786	128,905	131,816	133,739	135,256	136,439	137,341	137,903	139,082	140,813
2002	52,590	103,196	121,800	130,075	134,368	137,402	139,407	140,989	142,221	143,162	143,747	144,976	146,781
2003	58,599	112,342	133,543	142,616	147,323	150,650	152,848	154,582	155,933	156,965	157,606	158,954	160,933
2004	60,361	121,757	144,735	154,568	159,670	163,276	165,658	167,538	169,002	170,120	170,815	172,276	174,421

$121757 = 60361 * 2.017$

$\hat{C}_{i,j}$



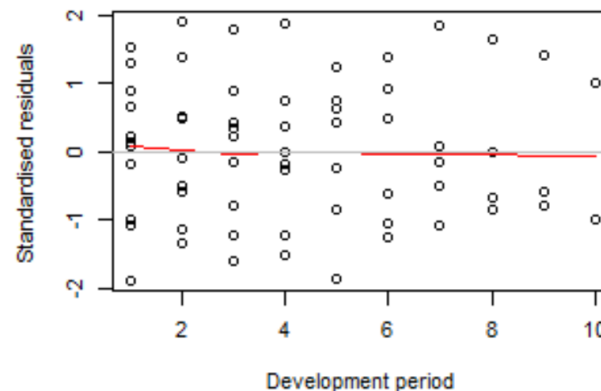
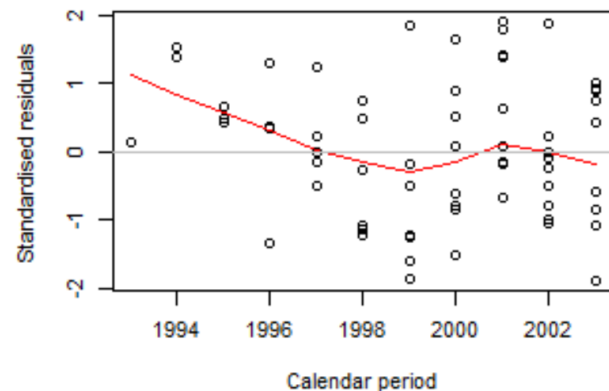
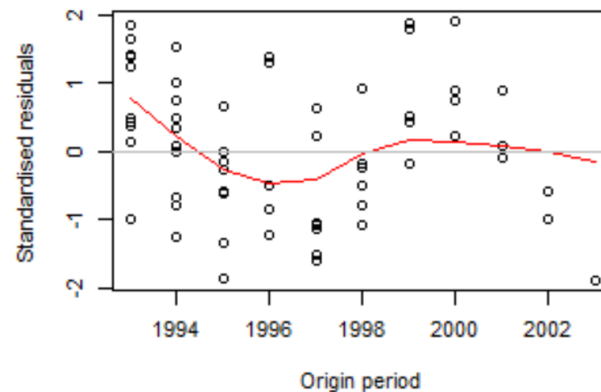
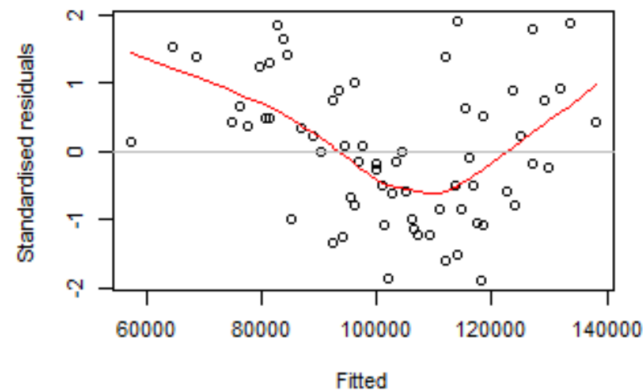
$$C_{i,n-i} \quad \hat{C}_{i,\infty} \quad R^{PCL}$$

	Latest Cum. Payments	Ultimate Cost	Claims Reserve
1993	85,774	86,843	1,068
1994	96,325	98,358	2,033
1995	105,008	107,662	2,655
1996	114,675	118,351	3,677
1997	118,289	123,149	4,860
1998	132,138	139,128	6,990
1999	138,291	147,731	9,440
2000	129,901	141,901	12,000
2001	124,786	140,813	16,027
2002	121,800	146,781	24,982
2003	112,342	160,933	48,591
2004	60,361	174,421	114,060
	1,339,690	1,586,073	246,383

Insurer Reserve	Adequacy
1,068	0
1,795	-238
1,853	-801
1,901	-1,776
2,845	-2,015
4,926	-2,064
7,718	-1,722
12,339	339
18,725	2,698
29,487	4,505
47,921	-670
107,822	-6,238
238,401	-7,982

Paid Chain-Ladder method shows a deficiency of roughly 8 mln of Euro (equal to 3.35% of the amount posted by the company).

Residuals



- One simple way to compute residuals of the CL model is:

$$y_{i,j} = \frac{C_{i,j} - C_{i,j-1}\hat{\lambda}_{j-1}}{C_{i,j-1}} = f_{i,j-1} - \hat{\lambda}_{j-1}$$
- Figures report these residuals, after that a classical standardization has been applied.
- These standardized residuals should not show any trends in any of the (time-) axes (accident year axis, development year axis and calendar year axis, as well as ordered by their sizes). The graphs may indicate that there are accident year and/or calendar year trends (which may also be induced by the increasing volume in the portfolio).
- Notice that different residuals have also been defined in literature. Some of them will be used in the next Sections.

2.3. Incurred Chain-Ladder Method

Incurred Losses

- Similarly to Paid Chain-Ladder, it has been developed an Incurred Chain-Ladder method.
- This methodology exploit loss development framework by using Incurred Losses.
- Incurred Losses are usually defined as:

$$I_{i,j} = C_{i,j} + R_{i,j}$$

- The term incurred losses is used by many throughout the P&C Industry. It can be misunderstood as to whether or not it includes IBNR. Many actuaries use the labels *case incurred* and *incurred on reported claims* to specifically note that the losses do not include IBNR. Furthermore, the incurred amount can be obtained by considering the case reserve or the booked one (hence including for instance provision for future development of known claims).

Incremental and Cumulative Payments

Outstanding Claims

 $R_{i,j}$

Amounts in thousand of Euro (.000)

	0	1	2	3	4	5	6	7	8	9	10	11
1993	56,551	26,907	12,910	7,836	6,201	4,555	3,806	3,189	2,792	1,987	1,784	1,068
1994	63,344	25,706	12,497	7,797	6,088	4,289	3,743	2,915	2,252	1,762	1,795	
1995	66,954	25,502	13,034	8,177	5,767	4,623	3,620	2,547	1,804	1,853		
1996	72,768	26,613	13,663	9,695	7,363	6,052	4,111	3,060	1,901			
1997	76,338	29,399	16,012	10,904	8,774	6,308	5,255	2,845				
1998	83,493	37,080	19,807	14,191	11,144	8,161	4,926					
1999	92,136	44,671	25,963	17,438	12,034	7,718						
2000	97,059	47,808	26,658	17,590	12,339							
2001	94,658	44,919	26,905	18,725								
2002	104,836	48,968	29,487									
2003	112,265	47,921										
2004	107,822											

Cumulative Paid Claims

 $C_{i,j}$

Amounts in thousand of Euro (.000)

	0	1	2	3	4	5	6	7	8	9	10	11
1993	28,446	57,697	70,161	75,306	78,032	80,392	81,726	82,964	83,906	84,766	85,047	85,775
1994	31,963	68,069	81,511	87,379	90,261	92,683	93,601	94,677	95,411	95,869	96,325	
1995	37,775	77,900	90,851	96,885	99,895	101,159	102,409	103,544	104,448	105,007		
1996	40,418	84,917	100,287	105,881	108,496	110,481	112,617	113,802	114,674			
1997	44,116	89,606	104,946	110,424	112,964	115,871	117,165	118,289				
1998	50,294	98,335	116,177	123,213	127,146	129,872	132,139					
1999	49,620	99,611	119,181	129,228	134,978	138,291						
2000	46,410	96,104	116,985	125,186	129,900							
2001	48,295	97,648	115,953	124,786								
2002	52,590	103,196	121,800									
2003	58,599	112,342										
2004	60,361											

Incurred Losses

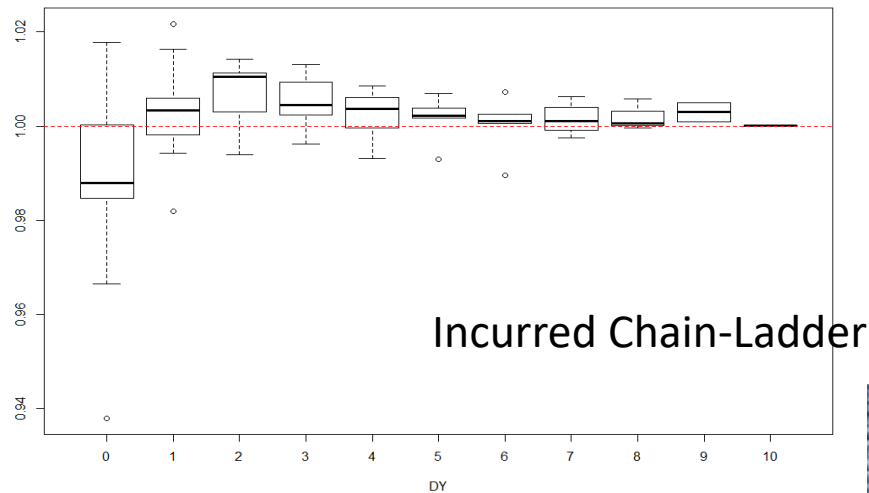
Incurred Losses

	0	1	2	3	4	5	6	7	8	9	10	11
1993	84,997	84,604	83,071	83,141	84,234	84,947	85,532	86,153	86,698	86,752	86,832	86,843
1994	95,307	93,775	94,008	95,176	96,349	96,972	97,344	97,592	97,662	97,631	98,120	
1995	104,728	103,402	103,885	105,062	105,662	105,782	106,029	106,091	106,252	106,861		
1996	113,185	111,529	113,950	115,576	115,859	116,533	116,729	116,862	116,576			
1997	120,454	119,006	120,958	121,328	121,739	122,179	122,420	121,135				
1998	133,787	135,414	135,985	137,404	138,290	138,033	137,064					
1999	141,756	144,283	145,144	146,667	147,012	146,009						
2000	143,469	143,913	143,643	142,776	142,240							
2001	142,952	142,567	142,857	143,511								
2002	157,426	152,164	151,287									
2003	170,864	160,263										
2004	168,183											

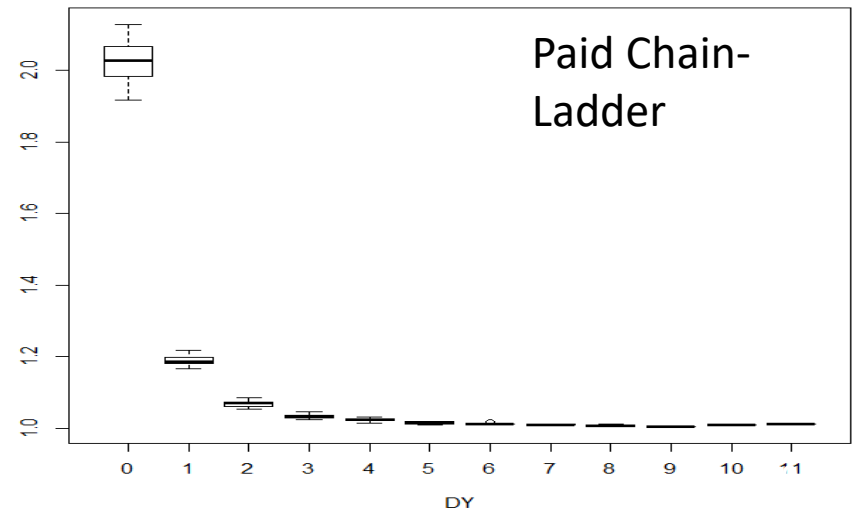
Individual loss development factors (age-to-age)

0	1	2	3	4	5	6	7	8	9	10
0.995	0.982	1.001	1.013	1.008	1.007	1.007	1.006	1.001	1.001	1.000
0.984	1.002	1.012	1.012	1.006	1.004	1.003	1.001	1.000	1.005	
0.987	1.005	1.011	1.006	1.001	1.002	1.001	1.002	1.006		
0.985	1.022	1.014	1.002	1.006	1.002	1.001	0.998			
0.988	1.016	1.003	1.003	1.004	1.002	0.989				
1.012	1.004	1.010	1.006	0.998	0.993					
1.018	1.006	1.010	1.002	0.993						
1.003	0.998	0.994	0.996							
0.997	1.002	1.005								
0.967	0.994									
0.938										

Individual Loss Development Factors



Individual Loss Development Factors



Development factors

As for Paid CL, the Incurred CL factors λ_j are not known and, henceforth, need to be estimated. A classical way is given by the following formula:

$$\hat{\lambda}_j^I = \frac{\sum_{i=0}^{n-j-1} I_{i,j+1}}{\sum_{i=0}^{n-j-1} I_{i,j}} = \frac{\sum_{i=0}^{n-j-1} f_{i,j}^I I_{i,j}}{\sum_{i=0}^{n-j-1} I_{i,j}}, \text{ for } j = 0, \dots, n-1$$

Hence, we have a row vector $\hat{\lambda}^I = [\hat{\lambda}_0^I, \dots, \hat{\lambda}_{n-1}^I]$.

Alternative estimators (as simple average or the average of only last 3,4,5 diagonals) are often used in practice.

Remarks:

- **A first estimation of tail development** is already included in the last development factor $\hat{\lambda}_{n-1}^I$.
- There is a strict relation between the value of $f_{i,j}^I$ and the run-off result of accident year i in calendar year $i + j + 1$.

When incurred losses are based on booked reserve, a value of $f_{i,j}^I > 1$ means that the estimation of ultimate cost (for AY i) increased from time $i + j$ to $i + j + 1$ (i.e. a negative run-off).

Lower Triangle and Incurred CL reserves

- Incurred amounts i, j of lower triangle are predicted as:

$$\hat{I}_{i,j} = I_{i,n-i} \prod_{h=n-i}^{j-1} \hat{\lambda}_h^I$$

for $i = 1, \dots, n$ $j = n - i + 1, \dots, n$

- The undiscounted claim reserve, derived by Incurred Chain-Ladder method, is:

$$R^{ICL} = \sum_{i=0}^n (\hat{I}_{i,n} - C_{i,n-i})$$

where $\hat{I}_{0,n} = I_{0,n}$ if no alternative estimation of tail is developed.

$$C_{i,n-i} \quad \hat{I}_{i,n} \quad R^{ICL}$$

	Latest Cum. Payments	Ultimate Cost	Claims Reserve
1993	85.774	86.842	1.068
1994	96.325	98.133	1.808
1995	105.008	107.205	2.197
1996	114.675	117.205	2.530
1997	118.289	121.935	3.645
1998	132.138	137.912	5.773
1999	138.291	147.061	8.770
2000	129.901	143.496	13.596
2001	124.786	145.429	20.643
2002	121.800	154.319	32.519
2003	112.342	164.024	51.682
2004	60.361	169.930	109.569
	1.339.690	1.593.490	253.799

Insurer Reserve	Adequacy
1.068	0
1.795	-13
1.853	-344
1.901	-629
2.845	-800
4.926	-847
7.718	-1.052
12.339	-1.257
18.725	-1.918
29.487	-3.032
47.921	-3.761
107.822	-1.747
238.400	-15.399

Incurred Chain-Ladder method shows a deficiency of roughly 15 mln of Euro (equal to 6.46% of the amount posted by the company).

2.4 Expected Loss Ratio

- Insurers often use expected claims techniques when entering new lines of business or new territories.
- Many actuaries use also this method for estimating unpaid claims for the most immature period(s).
- Expected loss ratios are a critical component of several other methods, as for instance Bornhuetter-Ferguson and Cape Cod (discussed later)
- The key assumption of the expected claims technique is that the actuary can better estimate total unpaid claims based on an a priori (or initial estimate) than from claims experience observed in the triangle. In some cases, claims experience can provide little information about ultimate claims (e.g. for start-up insurers)

- The aim is:
 - To quantify the exposure for each accident year
 - To quantify the expected ultimate loss ratio for each accident year
- There are several ways for actuaries to estimate the a-priori expected ultimate loss ratio. Main approaches are based on the use of the estimation developed for ratemaking or budget purposes (internal data). In some circumstances, external data (e.g. from the market) can be useful for this issue.

For further details, see
Chapter 8: *Expected Claims Technique (pag. 131-151)*,
Estimating Unpaid Claims Using Basic Techniques
Casualty Actuarial Society 2009

2.5 (Paid and Incurred) Bornhuetter-Ferguson

- Bornhuetter-Ferguson (BF) method (see *The Actuary and IBNR*, Bornhuetter R.L., Ferguson R.E., 1972) is developed as a mix of *development techniques* and *expected claims techniques*.
- Indeed, it can be seen as a credibility weighting between the development method and the expected claims method.
 - In the **development method** (as Paid Chain-Ladder), **full credibility** is given to actual claims experience
 - In the **expected claims method** (as Expected Loss Ratio), **no credibility** is given to claims experience.
- In Bornhuetter-Ferguson method, *credibility is equal to the percentage of claims developed at a particular stage of maturity*, that is a function of the cumulative claims development factor.
- Therefore, **more weight is given to the expected claims method in less mature years** and **more weight is given to the development method in more mature years of the experience period**.

PAID Bornhuetter-Ferguson

- The key assumptions are that:
 - unpaid claims will develop based on expected claims technique (do not depend on paid claims as in chain-ladder)
 - payment patterns are the same as those selected in chain-ladder.
- So it is based on following values:
 - A prior estimation $\hat{\mu}_i$ of the expected ultimate claim for each AY i: $\hat{\mu}_i = E(C_{i,\infty})$
 - The claims development patterns
- Paid BF assumes that the ultimate cost U_i^{BF} can be obtained as:
$$U_i^{BF} = C_{i,\infty}^{PCL}(z_i) + \hat{\mu}_i(1 - z_i)$$

with z_i ($i = 0, \dots, n$) is a credibility weight.

- It is assumed that $\hat{\mu}_i$ is given externally by expert opinion (do not depend on data).
 - For instance $\hat{\mu}_i = \widehat{ULR}_i \cdot B_i$
- Payment patterns are usually based on chain-ladder development factors. In this case, we can introduce the cumulative claims development factor as:

$$\widehat{CDF}_j = \prod_{h=j}^{n-1} \hat{\lambda}_h$$

- The product goes to n (or to last DY-1) in case of tail
- $1/\widehat{CDF}$ is a sort of cumulative settlement speed

- The weight is defined as

$$\hat{z}_i = \frac{1}{\widehat{CDF}_{n-i}}$$

- Under previous assumptions we have:

$$\begin{aligned}
 U_i^{BF} &= C_{i,\infty}^{PCL} \left(\frac{1}{\widehat{CDF}_{n-i}} \right) + \hat{\mu}_i \left(1 - \frac{1}{\widehat{CDF}_{n-i}} \right) = \\
 &= C_{i,n-i} + \hat{\mu}_i \left(1 - \frac{1}{\widehat{CDF}_{n-i}} \right)
 \end{aligned}$$



$$R_i^{BF} = \hat{\mu}_i \left(1 - \frac{1}{\widehat{CDF}_{n-i}} \right)$$

Notice that:

$$R_i^{PCL} = C_{i,\infty}^{PCL} - C_{i,n-i} = C_{i,\infty}^{PCL} \left(1 - \frac{1}{\widehat{CDF}_{n-i}} \right)$$

- For SIFA Insurer, we have:

j	0	1	2	3	4	5	6	7	8	9	10	11
Dev. Factors (PCL)	2.017	1.189	1.068	1.033	1.023	1.015	1.011	1.009	1.007	1.004	1.009	1.012
CDF	2.890	1.433	1.205	1.128	1.092	1.068	1.053	1.041	1.032	1.025	1.021	1.012
z_{t-j}	34.61%	69.81%	82.98%	88.62%	91.54%	93.61%	94.98%	96.05%	96.89%	97.53%	97.93%	98.77%

AY	$C_{i,t-i}$	CDF_i	U^{CL}	LR^{CL}	LR^{EC}	B	U^{EC}	z_i	U^{BF}	R^{BF}
1993	85,775	1.012	86,843	82.37%	82.46%	105,432	86,942	98.77%	86,844	1,069
1994	96,325	1.021	98,358	90.54%	90.24%	108,635	98,035	97.93%	98,352	2,027
1995	105,007	1.025	107,662	95.38%	94.84%	112,879	107,057	97.53%	107,647	2,640
1996	114,674	1.032	118,351	99.51%	98.53%	118,932	117,178	96.89%	118,315	3,641
1997	118,289	1.041	123,149	102.07%	101.27%	120,654	122,191	96.05%	123,112	4,822
1998	132,139	1.053	139,128	98.15%	97.25%	141,755	137,852	94.98%	139,064	6,926
1999	138,291	1.068	147,731	93.58%	93.34%	157,865	147,346	93.61%	147,706	9,415
2000	129,900	1.092	141,901	89.63%	90.99%	158,323	144,061	91.54%	142,083	12,183
2001	124,786	1.128	140,813	87.39%	90.45%	161,137	145,751	88.62%	141,375	16,589
2002	121,800	1.205	146,781	88.54%	93.36%	165,782	154,773	82.98%	148,141	26,342
2003	112,342	1.433	160,933	85.76%	87.39%	187,659	163,988	69.81%	161,856	49,514
2004	60,361	2.890	174,421	89.65%	85.82%	194,567	166,974	34.61%	169,551	109,190
Totale										244,356

Ultimate Cost,
Paid Chain-
Ladder

*a-priori expected
ultimate loss ratio*

a-priori Expected
Ultimate Cost

Incurred Bornhuetter-Ferguson

- The idea of Paid BF can be theoretically extended by using the Incurred method.
- We would have that the ultimate cost U_i^{IBF} is:

$$U_i^{IBF} = C_{i,n}^{ICL}(z_i) + \hat{\mu}_i(1 - z_i)$$

with $\hat{z}_i = \frac{1}{\widehat{CDF}_{n-i}^I} = \frac{1}{\prod_{h=n-i}^{n-1} \hat{\lambda}_h^I}, (i = 1, \dots, n).$

- Main problem of this approach is that $\hat{\lambda}_h^I$ is usually very close to 1 and we could have a weight $\hat{z}_i$ greater than one.

j	0	1	2	3	4	5	6	7	8	9	10	11
Dev. Factors (ICL)	0.987	1.003	1.007	1.004	1.002	1.001	1.000	1.001	1.002	1.003	1.000	1.000
CDF	1.010	1.023	1.020	1.013	1.009	1.007	1.006	1.007	1.005	1.003	1.000	1.000
z_{t-j}	98.97%	97.71%	98.04%	98.68%	99.12%	99.29%	99.39%	99.34%	99.46%	99.68%	99.99%	100.00%

AY	$l_{i,t}$	CDF	U^{ICL}	LR_{EC}	B	U_{EC}	z_i	U_{BF}	$C_{i,t}$	R_{BF}
1993	86,843	1.000	86,843	82.46%	105,432	86,942	100.00%	86,843	85,775	1,068
1994	98,120	1.000	98,133	90.35%	108,635	98,148	99.99%	98,133	96,325	1,808
1995	106,861	1.003	107,204	95.05%	112,879	107,286	99.68%	107,205	105,007	2,197
1996	116,576	1.005	117,205	99.01%	118,932	117,751	99.46%	117,208	114,674	2,533
1997	121,135	1.007	121,935	101.78%	120,654	122,798	99.34%	121,941	118,289	3,651
1998	137,064	1.006	137,912	97.68%	141,755	138,460	99.39%	137,915	132,139	5,777
1999	146,009	1.007	147,061	93.55%	157,865	147,681	99.29%	147,065	138,291	8,774
2000	142,240	1.009	143,496	90.49%	158,323	143,263	99.12%	143,494	129,900	13,594
2001	143,511	1.013	145,429	89.02%	161,137	143,443	98.68%	145,403	124,786	20,617
2002	151,287	1.020	154,319	91.09%	165,782	151,004	98.04%	154,254	121,800	32,454
2003	160,263	1.023	164,024	86.56%	187,659	162,443	97.71%	163,987	112,342	51,645
2004	168,183	1.010	169,930	86.97%	194,567	169,219	98.97%	169,923	60,361	109,561
Total										253,679

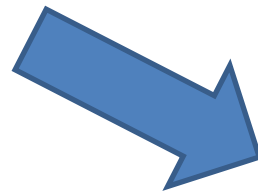
Some variants of BF Cape Cod and others

Expected ultimate loss ratio.

$$U_i^{CC} = C_{i,n-i} + B_i \frac{\sum_{i=0}^n C_{i,n-i}}{\sum_{i=0}^n B_i z_i} (1 - z_i)$$

Under Paid CL patterns, we have:

$$\hat{z}_i = \frac{1}{\widehat{CDF}_{n-i}}$$



$$U_i^{CC} = C_{i,n-i} + B_i \frac{\sum_{i=0}^n \frac{C_{i,n-i}}{B_i z_i} B_i z_i}{\sum_{i=0}^n B_i z_i} (1 - z_i) = C_{i,n-i} + B_i \frac{\sum_{i=0}^n ULR_i^{PCL} B_i z_i}{\sum_{i=0}^n B_i z_i} (1 - z_i)$$

Other extensions have been proposed in Berkander (1976) and Hovinen (1981).

For further details, see
Chapter 9 Bornhuetter-Ferguson Technique (pag. 152-173)
Estimating Unpaid Claims Using Basic Techniques
Casualty Actuarial Society 2009

2.6 Fisher-Lange Method

Fisher-Lange method

- Fisher-Lange method has been proposed by Fisher and Lange in 1974 (see “Loss Reserve Testing: A Report Year Approach”, CAS Proceedings – **1974**).
- It is an average cost method, where outstanding claims are obtained by deriving a separate estimation of frequency (number of future payments for existing claims) and severity (average paid cost).
- A proper variant of Fisher-Lange method has been proposed in 1983 (see R. Ottaviani – Convegno SIS **1983**). In the next, we will focus only on this proposal.

Steps of the model

- FL method allows for a separate description of the number and the cost of claims. Differently from Chain-Ladder, it is not based only on the aggregate payments $P_{i,j}$.

$$P_{i,j} = n_{i,j}m_{i,j}$$

- The incremental number of claims ($n_{i,j}$) is supposed to be related to the number of open (reserved) claims at the valuation date through two specific coefficients (that need to be estimated):
 - the **percentage of open claims** having a payment in future years
 - the **claim settlement speed**, representing the timing of the claims count payments estimated to have effectively a payment in the future.
- The expected average paid cost ($m_{i,j}$) is estimated on the basis of the observed triangles and of the assumption on future inflation rates.
- Ultimate claims are forecasted by multiplying the estimate of the ultimate number of claims by the estimate of the ultimate average value.

Percentage of open claims having a payment in future years

- At the valuation date, the percentage $aliqu_{i,j}$ is the portion of open claims having effectively a payment in the future, regarding implicitly either nil, reopened and IBNR claims:

$$aliqu_{i,j} = \frac{\sum_{h=j+1}^{n-i} n_{i,h} + r_{i,n-i}}{r_{i,j}}$$

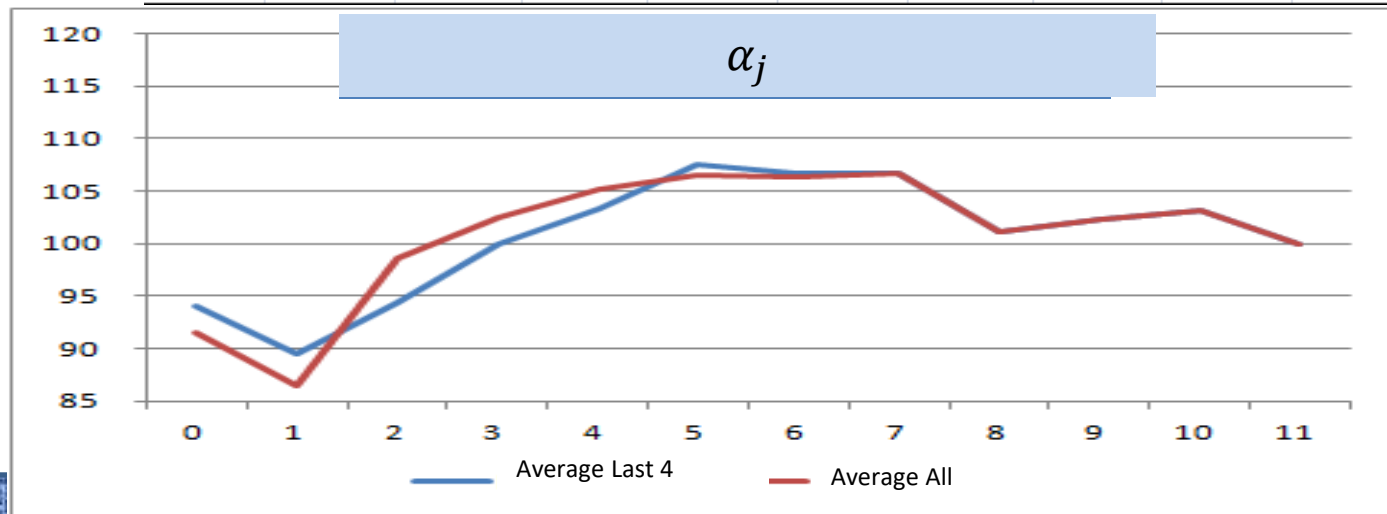
- $n_{i,j}$: **number of closed claims with payments** (AY i , DY j)
- $r_{i,j}$: **number of open claims** (AY i , DY j)

$$i = 0, \dots, n-1; j = 0, \dots, n-i-1$$

- A function of these percentages (usually the arithmetic mean for each development year) allows to obtain a vector of size n : $\alpha = [\alpha_j]_{j=0, \dots, n-1}$, where $\alpha_j = E(aliqu_{i,j})$. If a tail is needed (i.e. first accident year has open claims at the valuation date), a further estimation is needed for α_n (a common assumption is $\alpha_n = 1$)
- The expected total number of future paid claims of an accident year is derived by applying each percentage α_j to the number of open claims of last calendar year n ,

Percentage of open claims having a payment in future years

		DY					aliq _{i,j}							
AY		0	1	2	3	4	5	6	7	8	9	10	11	12+
1993		85.48	71.31	93.86	99.24	99.78	101.01	104.94	107.37	106.50	103.23	103.09	-	
1994		86.78	78.33	97.98	102.77	105.14	108.21	101.41	100.79	98.29	101.42	-		
1995		89.55	83.68	102.76	107.61	117.87	114.87	117.06	115.83	98.46	-			
1996		91.27	87.85	107.13	110.78	109.49	109.48	106.24	103.02	-				
1997		92.50	94.61	107.65	108.48	104.42	104.03	102.17	-					
1998		92.40	91.47	99.81	98.75	99.80	101.48	-						
1999		92.35	86.42	91.71	94.91	99.54	-							
2000		89.90	85.07	92.39	97.42	-								
2001		93.02	92.89	93.51	-									
2002		96.44	93.46	-										
2003		96.98	-											
2004		-												
α _j (*100)		91.52	86.51	98.53	102.50	105.15	106.51	106.37	106.75	101.08	102.33	103.09		
		91.52	86.51	98.53	102.50	105.15	106.51	106.37	106.75	101.08	102.33	103.09	100.00	



Claim settlement speed

- The claim settlement speed represents the timing of the claims count payments estimated to have effectively a payment in the future. In practice, we need an estimate concerning the timing of the paid claims count in the future for each separate accident year. For the last accident year ($i = n$), the claim settlement speed $v_j^{(n)}$ is computed as:

$$v_j^{(n)} = \frac{n_{n-j,j} \cdot \frac{d_n}{d_{n-j}}}{\sum_{j=1}^n n_{n-j,j} \cdot \frac{d_n}{d_{n-j}}}$$

- d_i : **number of reported claims** (AY i)
- When $i = n$, $v_j^{(n)}$ it is computed for $j = 1, \dots, n$
- If a tail is needed, then also $v_{n^+}^{(n)}$ must be computed. In this case, the summation at the denominator must be extended to n^+

- It is obtained by the number of claims effectively paid in the last calendar year for different development years, but where the origin of different accident years has to be made homogeneous (by a proportionality rule based on the number of reported claims d_i).
- For previous accident years** ($i < n$), the computation of $v_j^{(i)}$ is based on a similar formula neglecting the development years that are not needed. For instance if $i = n - 1$, then $v_j^{(n-1)}$ is computed for $j = 2, \dots, n$ and the summation will be from 2 to n (or n^+ in case of tail)

Claim settlement speed (an example)

Claim Settlement Speed for accident year $i=n$ and for each development year j (computed as in the formula of previous slide)

Estimation for the tail ($j=12^+$)

Selected settl. speed

10.955	1.646	627	264	239
78,30	11,77	4,48	1,89	1,71

Calendar Year n

10.955	1.646	627	264	239
78,30	11,77	4,48	1,89	1,71

Calendar Year n-1

10.339	1.874	681	325	144
76,26	13,82	5,02	2,40	1,06

Calendar Year n-2

10.552	1.732	616	224	90
78,74	12,92	4,59	1,67	0,68

Average of last 3 cal. years

77,77	12,84	4,70	1,98	1,15
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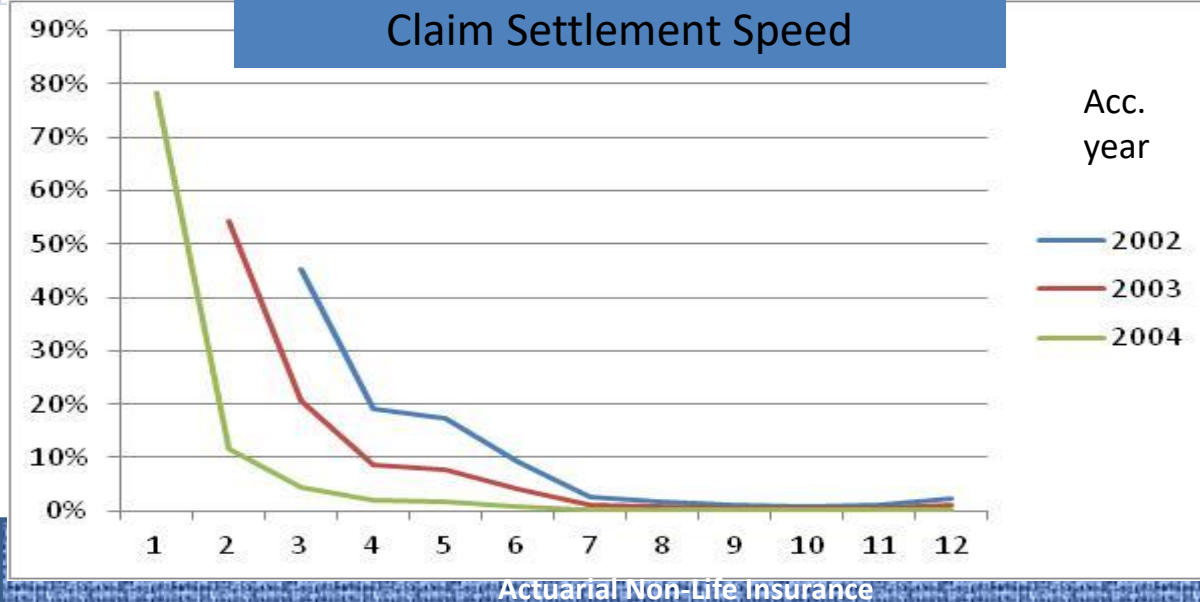
127	35	22	14	13	16	21
0,91	0,25	0,16	0,10	0,10	0,12	0,22
127	35	22	14	13	16	31
0,91	0,25	0,16	0,10	0,10	0,12	0,22
68	33	14	17	15	16	31
0,50	0,25	0,10	0,13	0,11	0,12	0,23
53	24	23	27	13	16	31
0,40	0,18	0,17	0,20	0,10	0,12	0,23
0,60	0,23	0,14	0,14	0,10	0,12	0,23

As a diagnostic tool, the claim settlement speed can be computed by replacing (in the formula of previous slide) values of last calendar with values of one of the previous diagonals

Expected number of future payments

i	$\hat{n}_{i,j}$														Totale	α_{n-i}	$r_{i,n-i}$
	0	1	2	3	4	5	6	7	8	9	10	11	12+				
1993	34,433	13,796	1,589	568	278	152	104	55	30	31	18	19	36	36	100.00	36	
1994	35,475	13,718	1,501	548	209	134	51	44	27	20	16	15	29	44	103.09	43	
1995	37,004	13,820	1,527	436	194	72	46	29	17	17	11	14	27	52	102.33	51	
1996	37,038	13,631	1,463	500	164	80	63	40	26	12	12	15	27	66	101.08	66	
1997	36,849	13,416	1,564	422	182	107	81	42	25	16	15	19	35	109	106.75	102	
1998	39,171	12,601	1,592	559	273	175	155	41	26	16	16	19	36	155	106.37	145	
1999	37,492	12,282	2,057	739	391	287	170	47	29	19	18	22	41	346	106.51	325	
2000	34,188	12,245	1,938	761	295	346	183	51	32	20	19	24	45	720	105.15	685	
2001	31,308	10,743	1,908	639	373	338	179	50	31	20	19	23	44	1,077	102.50	1,050	
2002	30,357	10,117	1,611	843	355	321	170	48	29	19	18	22	42	1,867	98.53	1,895	
2003	30,717	11,081	1,844	703	296	268	142	40	25	16	15	18	35	3,401	86.51	3,931	
2004	30,590	10,876	1,634	623	262	237	126	35	22	14	13	16	31	13,890	91.52	15,178	
Totale:														21,764	92.58%	23,508	

Claim Settlement Speed



$$\hat{n}_{i,j} = r_{i,n-i} \cdot \alpha_{n-i} \cdot v_j^{(i)}$$

$v_j^{(i)}$ Settlement speed for DY j and AY i

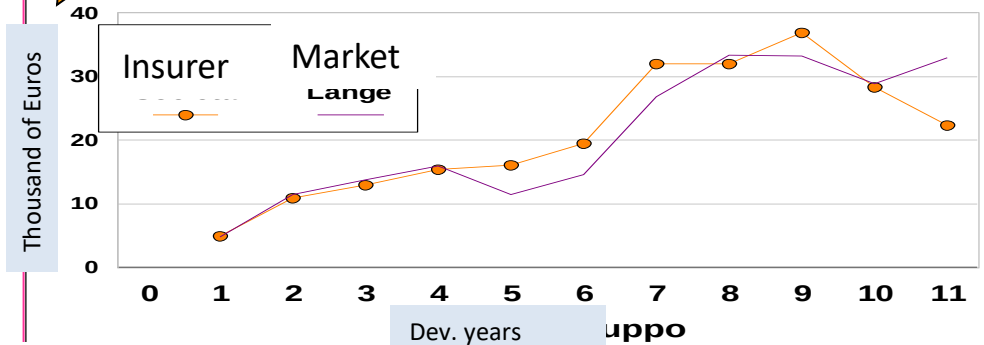
Average Paid Cost

To estimate expected average paid costs ($\hat{m}_{i,j}$) for each cell of the future triangle, we follow different steps:

1. we select for each development year the **average cost** (it could be the last observed value $m_{n-j,j}$, or a function of adjusted values of all/part of available average costs for the same duration (average by column));
2. we select the vector of **total claim inflation rates (both endogenous and exogenous), ir_h** , expected in the future (from 6% to 3% in the example);
3. **we inflate the chosen average cost** (equal to the last observed value in the formula) of each column, according to the total claim inflation rates:

$$\hat{m}_{i,j} = m_{n-j,j} \prod_{h=n+1}^{i+j} (1 + ir_h)$$

Average Cost (Insurer vs Market)



Expected Claim Inflation rates for future calendar years

Anno natam.	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016
exog. infl.	1,025	1,023	1,020	1,018	1,015	1,015	1,015	1,015	1,015	1,015	1,015	1,015
endog. infl.	1,035	1,030	1,025	1,020	1,015	1,015	1,015	1,015	1,015	1,015	1,015	1,015
	1,060	1,053	1,045	1,038	1,030	1,030	1,030	1,030	1,030	1,030	1,030	1,030

	$\hat{m}_{i,j}$ (in EURO)											
j	0	1	2	3	4	5	6	7	8	9	10	11
i												
1993												33.054
1994											28.884	35.037
1995										33.245	30.617	36.877
1996									33.458	35.240	32.224	38.536
1997								26.902	35.465	37.090	33.674	39.981
1998							14.654	28.516	37.327	38.759	34.937	41.181
						11.542	15.533	30.014	39.007	40.213	35.985	42.416
					15.969	12.235	16.349	31.364	40.470	41.419	37.065	43.689
2001				13.828	16.927	12.877	17.085	32.540	41.684	42.662	38.176	44.999
2002			11.548	14.657	17.816	13.457	17.725	33.516	42.934	43.941	39.322	46.349
2003		4.850	12.240	15.427	18.618	13.961	18.257	34.522	44.222	45.260	40.501	47.740
2004		5.141	12.883	16.121	19.316	14.380	18.805	35.558	45.549	46.617	41.716	49.172

Incremental Payments and Claims Reserve

	$\hat{P}_{i,j}$													Thousand of Euros
	0	1	2	3	4	5	6	7	8	9	10	11	12+	
1993	28,446	29,251	12,464	5,144	2,727	2,359	1,334	1,238	941	860	282	727	1,266	
1994	31,963	36,106	13,441	5,868	2,882	2,422	918	1,076	734	458	456	542	1,070	
1995	37,775	40,125	12,951	6,034	3,010	1,264	1,250	1,135	904	559	352	524	1,027	
1996	40,418	44,499	15,370	5,594	2,616	1,984	2,137	1,184	873	437	381	563	1,096	
1997	44,116	45,490	15,339	5,478	2,541	2,906	1,294	1,124	878	587	508	745	1,440	
1998	50,294	48,040	17,843	7,035	3,934	2,726	2,267	1,181	957	636	546	795	1,537	
1999	49,620	49,991	19,570	10,047	5,750	3,313	2,633	1,421	1,144	754	643	936	1,809	
2000	46,410	49,694	20,881	8,202	4,714	4,229	2,996	1,606	1,283	840	716	1,042	2,015	
2001	48,295	49,354	18,304	8,833	6,313	4,351	3,060	1,628	1,291	846	721	1,049	2,028	
2002	52,590	50,606	18,604	12,361	6,318	4,323	3,018	1,595	1,265	828	706	1,028	1,986	
2003	58,599	53,743	22,574	10,844	5,503	3,738	2,591	1,369	1,086	711	606	882	1,705	
2004	60,361	55,913	21,057	10,043	5,060	3,412	2,365	1,250	991	649	553	805	1,557	

We obtained with **Fisher-Lange (FL)** an outstanding claim reserve equal to **€ 251.1 millions**. Booked reserve is instead equal to **€ 238.4 millions**

Hence, according to FL method, we observe a deficiency of **€ 12.7 mln (equal to 5.3% of booked reserve)**.

Estimated Claims Reserve	Booked Reserve	Difference (Booked – Estimate)
1,266	1,068	-198
1,612	1,795	183
1,903	1,853	-49
2,477	1,901	-575
4,158	2,845	-1,313
5,653	4,926	-727
9,341	7,718	-1,623
14,726	12,339	-2,387
21,287	18,725	-2,562
33,427	29,487	-3,940
51,611	47,921	-3,690
103,656	107,822	4,166
251,117	238,401	-12,715

Ratio: Difference / Booked

-5,33%

Incremental Payments, Ultimate Cost and Loss Ratio

$\hat{P}_{i,j}$														Thousand of Euros	
	0	1	2	3	4	5	6	7	8	9	10	11	12+	Accident Year	Ultimate Cost
1993	28,446	29,251	12,464	5,144	2,727	2,359	1,334	1,238	941	860	282	727	1,266	1993	87,041
1994	31,963	36,106	13,441	5,868	2,882	2,422	918	1,076	734	458	456	542	1,070	1994	97,937
1995	37,775	40,125	12,951	6,034	3,010	1,264	1,250	1,135	904	559	352	524	1,027	1995	106,910
1996	40,418	44,499	15,370	5,594	2,616	1,984	2,137	1,184	873	437	381	563	1,096	1996	117,151
1997	44,116	45,490	15,339	5,478	2,541	2,906	1,294	1,124	878	587	508	745	1,440	1997	122,447
1998	50,294	48,040	17,843	7,035	3,934	2,726	2,267	1,181	957	636	546	795	1,537	1998	137,792
1999	49,620	49,991	19,570	10,047	5,750	3,313	2,633	1,421	1,144	754	643	936	1,809	1999	147,632
2000	46,410	49,694	20,881	8,202	4,714	4,229	2,996	1,606	1,283	840	716	1,042	2,015	2000	144,626
2001	48,295	49,354	18,304	8,833	6,313	4,351	3,060	1,628	1,291	846	721	1,049	2,028	2001	146,073
2002	52,590	50,606	18,604	12,361	6,318	4,323	3,018	1,595	1,265	828	706	1,028	1,986	2002	155,227
2003	58,599	53,743	22,574	10,844	5,503	3,738	2,591	1,369	1,086	711	606	882	1,705	2003	163,953
2004	60,361	55,913	21,057	10,043	5,060	3,412	2,365	1,250	991	649	553	805	1,557	2004	164,017
														TOTALE	1,590,807

Accident Year	Earned Premiums	Ultimate Cost	Ultimate Cost / Earned Premiums
1993	105,432	87,041	82.6%
1994	108,635	97,937	90.2%
1995	112,879	106,910	94.7%
1996	118,932	117,151	98.5%
1997	120,654	122,447	101.5%
1998	141,755	137,792	97.2%
1999	157,865	147,632	93.5%
2000	158,323	144,626	91.3%
2001	161,137	146,073	90.7%
2002	165,782	155,227	93.6%
2003	187,659	163,953	87.4%
2004	194,567	164,017	84.3%

The ratio of ultimate cost to earned premiums allows to obtain a loss ratio for each accident year.

- In our example, there is a trend towards a reduction in 2003 and 2004 (**87.4%** and **84.3%** respectively).

2.7 Frequency Severity methods

- Frequency-severity methods allow to estimate claims reserve by separately taking account of frequency and severity. In other words, it is an average cost method (as also Fisher-Lange).
- There are different ways to estimate both components (claim count and average costs).
- A classic methodology is based on the application of classical chain-ladder method to both cumulative paid claims counts and cumulative average paid costs.

Cumulative data

Incremental payments

$$P_{i,j} = n_{i,j}m_{i,j}$$

Cumulative payments

$$C_{i,j} = n_{i,j}^c m_{i,j}^c$$



$$n_{i,j}^c = \begin{cases} n_{i,j} & \text{if } j = 0 \\ n_{i,j} + n_{i,j-1}^c & \text{otherwise} \end{cases}$$

$$m_{i,j}^c = \frac{C_{i,j}}{n_{i,j}^c}$$

Cumulative Number of claims

	0	1	2	3	4	5	6	7	8	9	10	11	12+
1993	34,433	48,229	49,819	50,387	50,665	50,816	50,920	50,975	51,004	51,036	51,053	51,072	51,109
1994	35,475	49,193	50,693	51,241	51,451	51,584	51,635	51,679	51,706	51,726	51,742		
1995	37,004	50,824	52,351	52,788	52,981	53,053	53,099	53,128	53,145	53,162			
1996	37,038	50,669	52,131	52,631	52,795	52,875	52,938	52,977	53,003				
1997	36,849	50,265	51,828	52,251	52,432	52,539	52,620	52,662					
1998	39,171	51,772	53,364	53,924	54,197	54,372	54,527						
1999	37,492	49,774	51,831	52,571	52,961	53,248							
2000	34,188	46,433	48,371	49,132	49,427								
2001	31,308	42,051	43,958	44,597									
2002	30,357	40,474	42,085										
2003	30,717	41,799											
2004	30,590												

Cumulative Average Costs

	0	1	2	3	4	5	6	7	8	9	10	11	12+
1993	826	1,196	1,408	1,495	1,540	1,582	1,605	1,628	1,645	1,661	1,666	1,679	1,699
1994	901	1,384	1,608	1,705	1,754	1,797	1,813	1,832	1,845	1,853	1,862		
1995	1,021	1,533	1,735	1,835	1,885	1,907	1,929	1,949	1,965	1,975			
1996	1,091	1,676	1,924	2,012	2,055	2,089	2,127	2,148	2,164				
1997	1,197	1,783	2,025	2,113	2,154	2,205	2,227	2,246					
1998	1,284	1,899	2,177	2,285	2,346	2,389	2,423						
1999	1,323	2,001	2,299	2,458	2,549	2,597							
2000	1,358	2,070	2,419	2,548	2,628								
2001	1,543	2,322	2,638	2,798									
2002	1,732	2,550	2,894										
2003	1,908	2,688											
2004	1,973												

Development factors

By applying Chain-Ladder idea, we can obtain:

$$\hat{\lambda}_j^n = \frac{\sum_{i=0}^{n-j-1} n_{i,j+1}^c}{\sum_{i=0}^{n-j-1} n_{i,j}^c}, \text{ for } j = 0, \dots, n-1$$

$$\hat{\lambda}_j^m = \frac{\sum_{i=0}^{n-j-1} m_{i,j+1}^c}{\sum_{i=0}^{n-j-1} m_{i,j}^c}, \text{ for } j = 0, \dots, n-1$$

Notice that two tail factors need to be estimated (if first accident year is open)

j	0	1	2	3	4	5	6	7	8	9	10	11
Dev factors (based on cumulative claims counts)	1.358	1.035	1.011	1.005	1.003	1.002	1.001	1.000	1.000	1.000	1.000	1.001
CDF	1.439	1.060	1.024	1.012	1.007	1.005	1.003	1.002	1.002	1.001	1.001	1.001
Dev factors (based on cumulative average costs)	1.488	1.147	1.056	1.028	1.020	1.013	1.011	1.008	1.006	1.004	1.008	1.012
CDF	2.009	1.350	1.177	1.115	1.084	1.063	1.050	1.039	1.030	1.024	1.020	1.012
Dev factors (Paid Chain-Ladder)	2.017	1.189	1.068	1.033	1.023	1.015	1.011	1.009	1.007	1.004	1.009	1.012
CDF	2.890	1.433	1.205	1.128	1.092	1.068	1.053	1.041	1.032	1.025	1.021	1.012

As in CHL, lower triangles of cumulative claim counts and average costs can be estimated by making use of development factors.

Claims reserve is obtained as:

$$R^{FS} = \sum_{i=0}^n (\hat{n}_{i,\infty}^c \hat{m}_{i,\infty}^c - C_{i,n-i})$$



Ultimate cost frequency-severity (AY i)

Cumulative numbers

	0	1	2	3	4	5	6	7	8	9	10	11	12+
1993	34.433	48.229	49.819	50.387	50.665	50.816	50.920	50.975	51.004	51.036	51.053	51.072	51.109
1994	35.475	49.193	50.693	51.241	51.451	51.584	51.635	51.679	51.706	51.726	51.742	51.761	51.798
1995	37.004	50.824	52.351	52.788	52.981	53.053	53.099	53.128	53.145	53.162	53.179	53.199	53.237
1996	37.038	50.669	52.131	52.631	52.795	52.875	52.938	52.977	53.003	53.027	53.044	53.064	53.101
1997	36.849	50.265	51.828	52.251	52.432	52.539	52.620	52.662	52.687	52.710	52.727	52.747	52.784
1998	39.171	51.772	53.364	53.924	54.197	54.372	54.527	54.570	54.596	54.620	54.638	54.659	54.697
1999	37.492	49.774	51.831	52.571	52.961	53.248	53.332	53.375	53.400	53.424	53.441	53.461	53.499
2000	34.188	46.433	48.371	49.132	49.427	49.563	49.641	49.681	49.704	49.726	49.742	49.761	49.796
2001	31.308	42.051	43.958	44.597	44.811	44.933	45.004	45.041	45.062	45.082	45.096	45.113	45.145
2002	30.357	40.474	42.085	42.564	42.768	42.885	42.953	42.987	43.008	43.027	43.041	43.057	43.087
2003	30.717	41.799	43.258	43.751	43.960	44.080	44.150	44.185	44.206	44.226	44.240	44.257	44.288
2004	30.590	41.538	42.988	43.478	43.686	43.805	43.875	43.910	43.931	43.950	43.964	43.981	44.012

Cumulative Average Costs

	0	1	2	3	4	5	6	7	8	9	10	11	12+
1993	826	1.196	1.408	1.495	1.540	1.582	1.605	1.628	1.645	1.661	1.666	1.679	1.699
1994	901	1.384	1.608	1.705	1.754	1.797	1.813	1.832	1.845	1.853	1.862	1.877	1.899
1995	1.021	1.533	1.735	1.835	1.885	1.907	1.929	1.949	1.965	1.975	1.983	1.999	2.022
1996	1.091	1.676	1.924	2.012	2.055	2.089	2.127	2.148	2.164	2.177	2.185	2.203	2.229
1997	1.197	1.783	2.025	2.113	2.154	2.205	2.227	2.246	2.265	2.279	2.287	2.306	2.333
1998	1.284	1.899	2.177	2.285	2.346	2.389	2.423	2.449	2.469	2.485	2.494	2.514	2.544
1999	1.323	2.001	2.299	2.458	2.549	2.597	2.631	2.658	2.680	2.697	2.707	2.729	2.761
2000	1.358	2.070	2.419	2.548	2.628	2.680	2.715	2.743	2.766	2.783	2.794	2.816	2.850
2001	1.543	2.322	2.638	2.798	2.876	2.933	2.971	3.003	3.027	3.046	3.058	3.083	3.119
2002	1.732	2.550	2.894	3.055	3.141	3.203	3.244	3.279	3.306	3.326	3.339	3.366	3.406
2003	1.908	2.688	3.084	3.256	3.347	3.413	3.457	3.494	3.523	3.544	3.558	3.587	3.629
2004	1.973	2.936	3.368	3.556	3.656	3.728	3.776	3.816	3.847	3.871	3.886	3.918	3.964

	Latest Cum. Payments	Ultimate Cum. Numbers	Ultimate cum. Av. Cost	Ultimate Cost	Claims Reserve
1993	85,774	51,109	1,699	86,843	1,068
1994	96,325	51,798	1,899	98,358	2,033
1995	105,008	53,237	2,022	107,662	2,654
1996	114,675	53,101	2,229	118,355	3,681
1997	118,289	52,784	2,333	123,155	4,866
1998	132,138	54,697	2,544	139,139	7,001
1999	138,291	53,499	2,761	147,734	9,443
2000	129,901	49,796	2,850	141,897	11,996
2001	124,786	45,145	3,119	140,799	16,013
2002	121,800	43,087	3,406	146,743	24,943
2003	112,342	44,288	3,629	160,717	48,375
2004	60,361	44,012	3,964	174,445	114,084
	1,339,690	596,554		1,585,847	246,157

Insurer Reserve	Adequacy
1,068	0
1,795	-238
1,853	-801
1,901	-1,779
2,845	-2,021
4,926	-2,075
7,718	-1,725
12,339	343
18,725	2,712
29,487	4,544
47,921	-454
107,822	-6,262
238,400	-7,756

Payments and claims
reserve are in thousand
of Euros

For further details, see
Chapter 11 Frequency-Severity Techniques (pag. 196-264),
Estimating Unpaid Claims Using Basic Techniques
Casualty Actuarial Society 2009

2.8 Summary of results

A summary of results of the different methodologies

Ultimate Cost

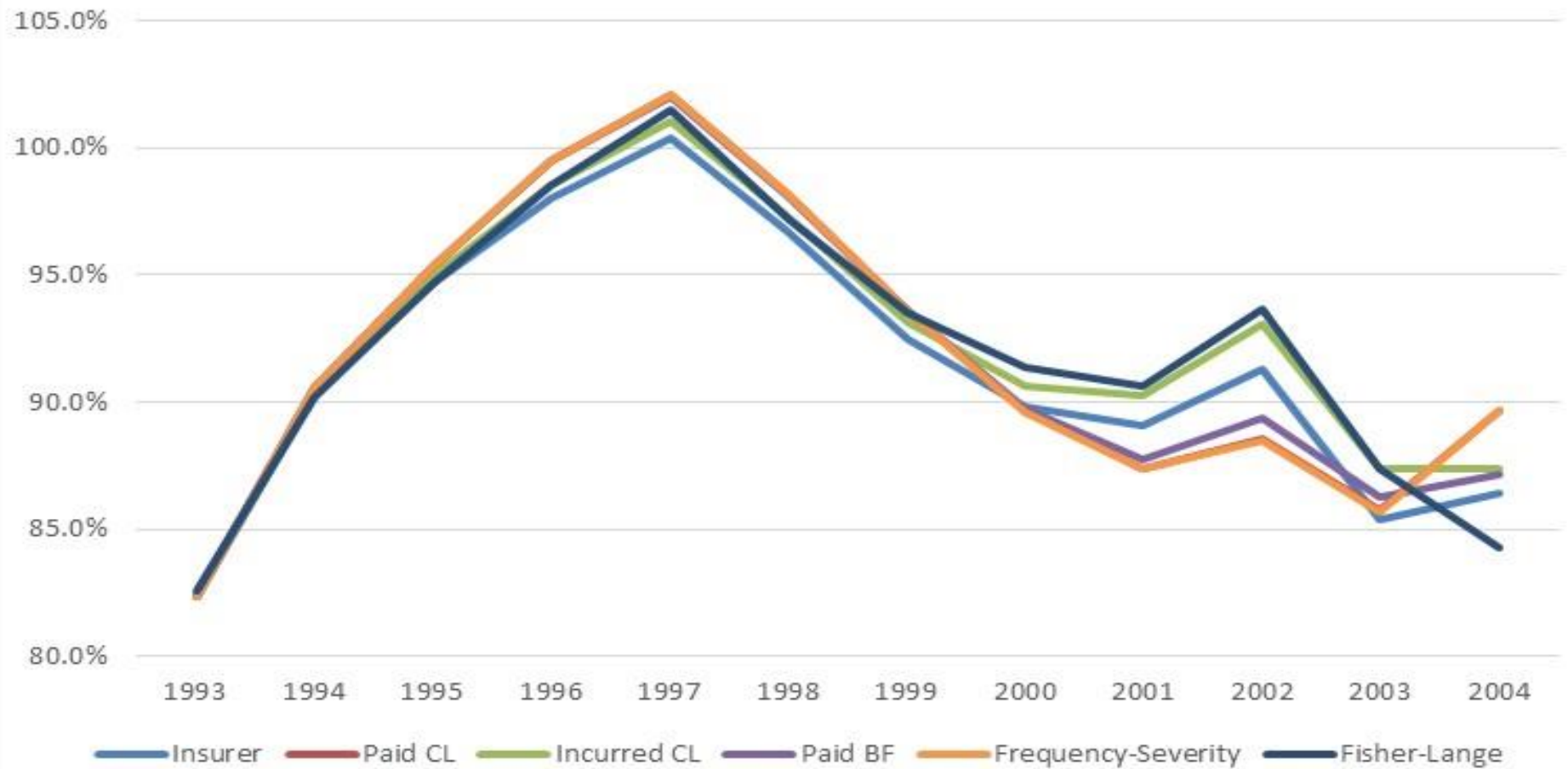
	Insurer	Paid CL	Incurred CL	Paid BF	Incurred BF	Fisher-Lange	Frequency-Severity
1993	86,843	86,843	86,843	86,844	86,843	87,041	86,843
1994	98,120	98,358	98,133	98,352	98,133	97,937	98,358
1995	106,861	107,662	107,204	107,647	107,205	106,910	107,662
1996	116,576	118,351	117,205	118,315	117,208	117,151	118,355
1997	121,135	123,149	121,935	123,112	121,941	122,447	123,155
1998	137,064	139,128	137,912	139,064	137,915	137,792	139,139
1999	146,009	147,731	147,061	147,706	147,065	147,632	147,734
2000	142,240	141,901	143,496	142,083	143,494	144,626	141,897
2001	143,511	140,813	145,429	141,375	145,403	146,073	140,799
2002	151,287	146,781	154,319	148,141	154,254	155,227	146,743
2003	160,263	160,933	164,024	161,856	163,987	163,953	160,717
2004	168,183	174,421	169,930	169,551	169,923	164,017	174,445

$$\text{Ultimate Loss Ratio} = \frac{\text{Ultimate Cost}}{\text{Earned Premiums}}$$

Ultimate Loss Ratio

	Earned Premiums	Insurer	Paid CL	Incurred CL	Paid BF	Incurred BF	Fisher-Lange	Frequency-Severity	Expected LR
1993	105,432	82.4%	82.4%	82.4%	82.4%	82.4%	82.6%	82.4%	82.5%
1994	108,635	90.3%	90.5%	90.3%	90.5%	90.3%	90.2%	90.5%	90.2%
1995	112,879	94.7%	95.4%	95.0%	95.4%	95.0%	94.7%	95.4%	94.8%
1996	118,932	98.0%	99.5%	98.5%	99.5%	98.6%	98.5%	99.5%	98.5%
1997	120,654	100.4%	102.1%	101.1%	102.0%	101.1%	101.5%	102.1%	101.3%
1998	141,755	96.7%	98.1%	97.3%	98.1%	97.3%	97.2%	98.2%	97.2%
1999	157,865	92.5%	93.6%	93.2%	93.6%	93.2%	93.5%	93.6%	93.3%
2000	158,323	89.8%	89.6%	90.6%	89.7%	90.6%	91.3%	89.6%	91.0%
2001	161,137	89.1%	87.4%	90.3%	87.7%	90.2%	90.7%	87.4%	90.5%
2002	165,782	91.3%	88.5%	93.1%	89.4%	93.0%	93.6%	88.5%	93.4%
2003	187,659	85.4%	85.8%	87.4%	86.2%	87.4%	87.4%	85.6%	87.4%
2004	194,567	86.4%	89.6%	87.3%	87.1%	87.3%	84.3%	89.7%	85.8%

ULR for accident year



3. Market Consistent Valuation of technical liabilities (Solvency II Framework)

Asset and liabilities

- ❑ Solvency II Directive defines a **market consistent valuation of assets and liabilities**:
 - **Assets** should be valued at the amount for which they *could be exchanged* between knowledgeable willing parties in an arm's length transaction;
 - **Liabilities** should be valued at the amount for which they *could be transferred*, or settled, between knowledgeable willing parties in an arm's length transaction.



- ❑ *“Consequently, the value of technical provisions should correspond to the amount another insurance or reinsurance undertaking (reference undertaking) would be expected to require to take over and meet the underlying insurance and reinsurance obligations” (Art.75 and following of the Solvency II Directive).*

Hedgeable and Non-Hedgeable

LIABILITIES

Hedgeable



Where future cash-flows associated with insurance or reinsurance obligations **can be replicated reliably using financial instruments** for which a reliable market value is observable, the value of technical provisions associated with those future cash-flows shall be determined on the basis of the market value of those financial instruments.
(i.e. Index and Unit Linked without guarantees)

Non-Hedgeable



Otherwise, a separate calculation of the best estimate and the risk margin shall be required. **The reserve will be the sum of:**

Best Estimate

+

Risk Margin

*"The best estimate should correspond to **the probability weighted average** of future cash-flows **taking account of the time value of money**" (Art. 77).*

*"The risk margin should be calculated by determining the **cost of providing an amount of eligible own funds equal to the SCR** necessary to support the insurance and reinsurance obligations **over the lifetime thereof**" (Art. 77).*

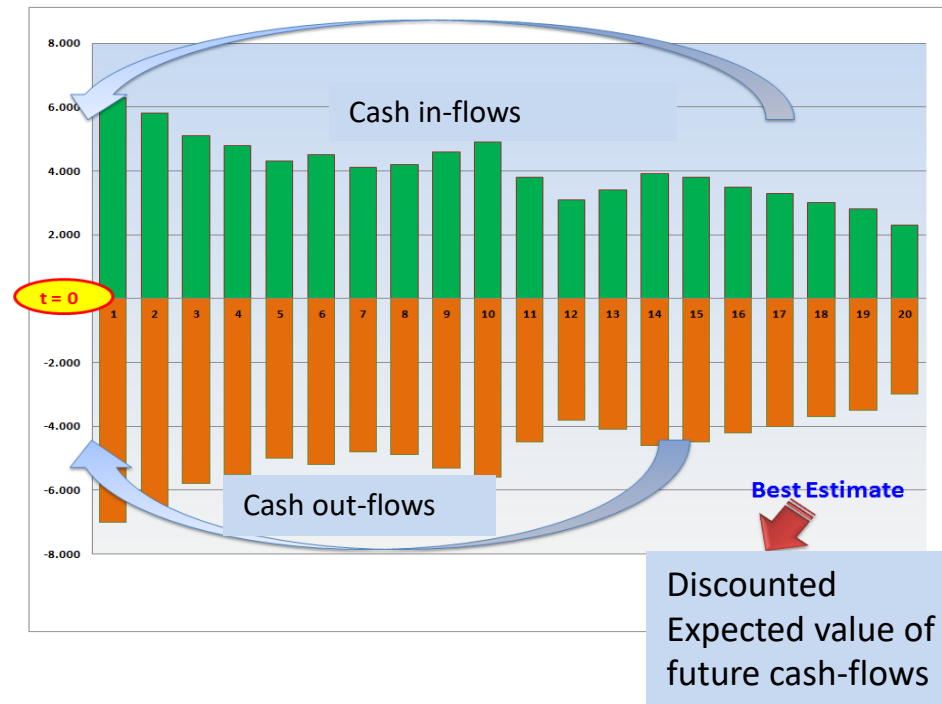
3.1. Best Estimate

Best Estimate (BE)

Some general comments

- Best Estimate is calculated on the basis of current and credible information and realistic assumptions, taking account of financial guarantees and options in insurance or reinsurance contracts, to deliver an economic valuation of insurance or reinsurance obligations.
- The cash-flow projection used in the calculation of the best estimate shall take account of all the cash in- and out-flows required to settle the insurance and reinsurance obligations over the lifetime thereof.
- Best estimate must be evaluated analitically for single claim or contract. It is allowed to reduce undue burden on the undertaking: the policies could be grouped and the projection of future cash-flows based on suitable model points could be permitted (it can be demonstrated that the grouping does not misrepresent the underlying risk and does not significantly misstate the costs).

- Best Estimate (BE) must be also evaluated net of reinsurance. $BE^{net} = BE^{gross} - \text{recoverables}$.
- Recoverables from reinsurance contracts shall take account of expected losses due to default of the counterparty. The adjustment for counterparty default should approximate the loss-given default of the counterparty, weighted with the probability of default of the counterparty.

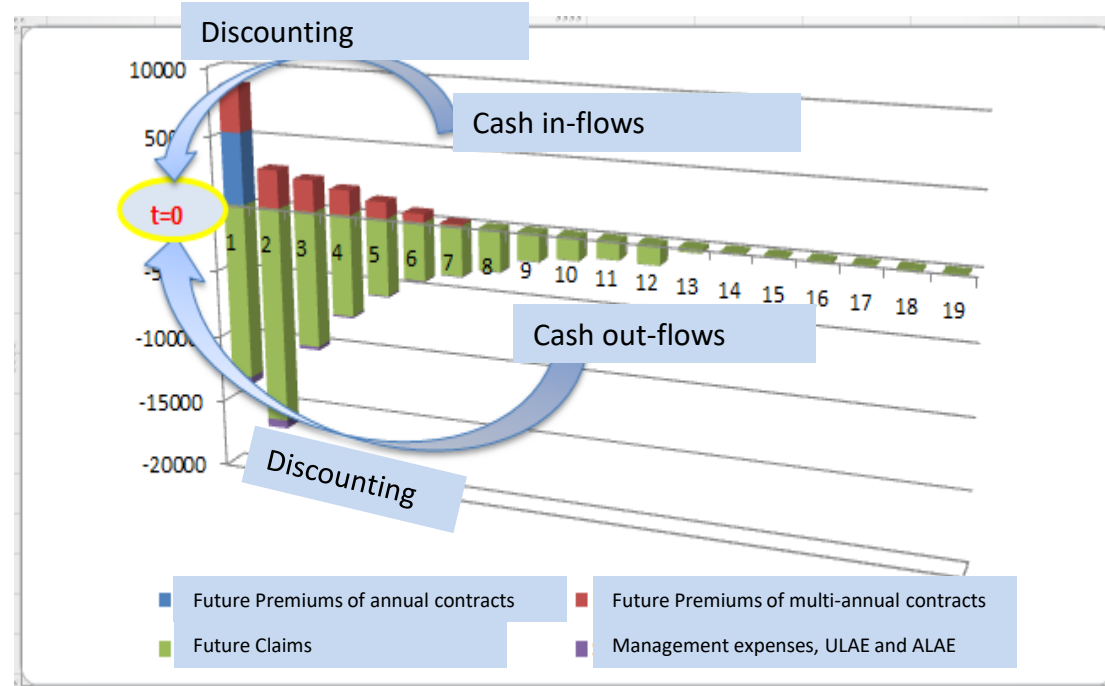


Best Estimate – Premium Reserve

- ❑ BE of Premium Reserve (evaluated separately from claims Reserve) takes into account both the **discounted expected value of future claims, management expenses and future premiums** for unexpired business at the valuation date.
- ❑ Premium Reserve could be lower of unearned premium reserves (UPR)

Main cash-flows (gross reinsurance)

- **Out-**
 - **Payments for future claims** incurred after the valuation date covered by existing business
 - **ALAE and ULAE for future claims** incurred after the valuation date covered by existing business
 - **Management costs** after the valuation date for existing business
- **In-**
 - **Future Premiums (net of acquisition costs) of existing annual contracts**
 - **Future Premiums (net of acquisition costs) of existing multi-annual contracts.** It should be taken into account the effect of lapses.



Simplification for BE of Premium Provision

$$BE = CR \cdot VM + (CR - 1)PVFP + AER \cdot PVFP$$

CR = estimate of **the combined ratio** for the line of business during the run-off period of the premium provision.

CR on a gross of acquisition cost basis i.e. $CR = (\text{incurred claims} + \text{claim related expenses} + \text{expenses}) / (\text{earned premiums gross of acquisition expenses})$.

Incurred claims should exclude the run-off result

Alternatively, if it is more practicable, the combined ratio for an accident year may be considered to be the sum of the expense ratio and the claims ratio (*ratio of the ultimate loss of incurred claims over earned premiums*).

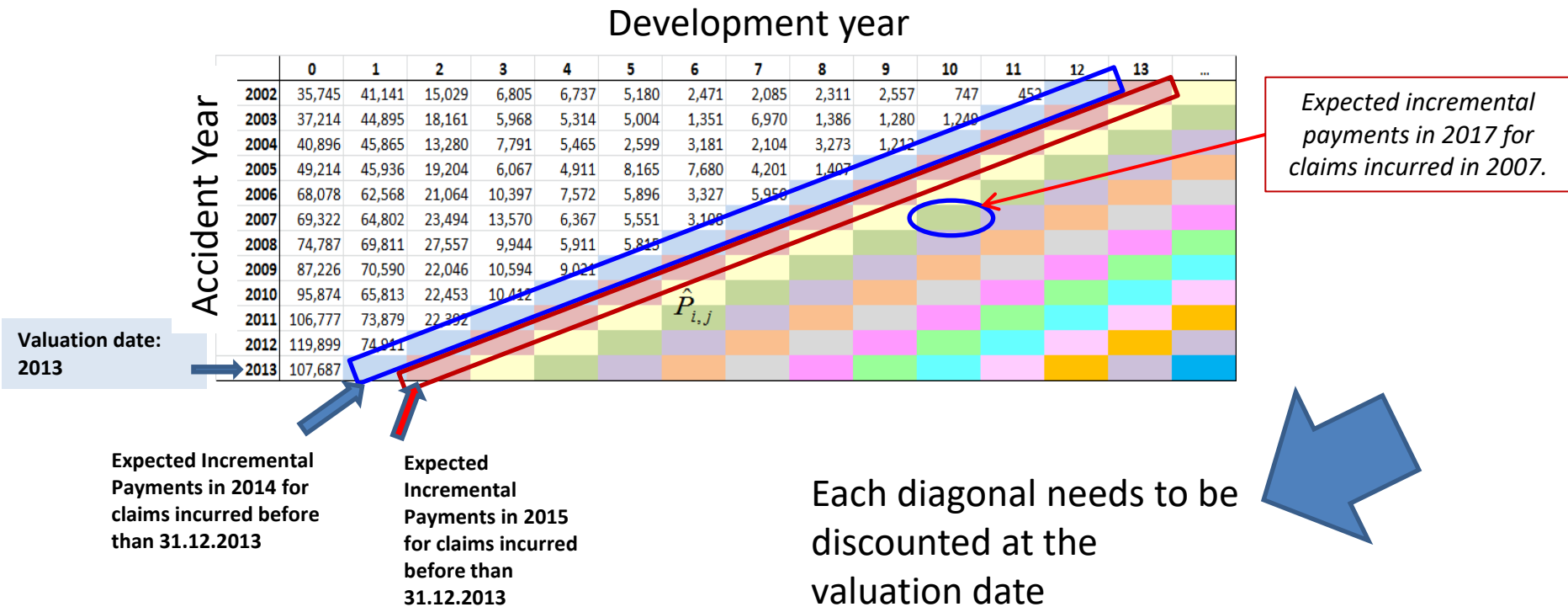
VM = **volume measure for unearned premium**. It relates to business that has incepted at the valuation date and represents the premiums for this incepted business less the premium that has already been earned against these contracts. This measure should be calculated gross of acquisition expenses.

PVFP = **present value of future premiums** (discounted using the prescribed term structure of risk-free interest rates) gross of commission.

AER = **estimate of acquisition expenses ratio** for line of business.

Best Estimate – Claims Reserve

- BE of claims reserves is derived as the discounted expected value of payments and expenses (as ULAE and ALAE) for claims already incurred at the valuation date (including IBNR). It should be evaluated net of recoverables.

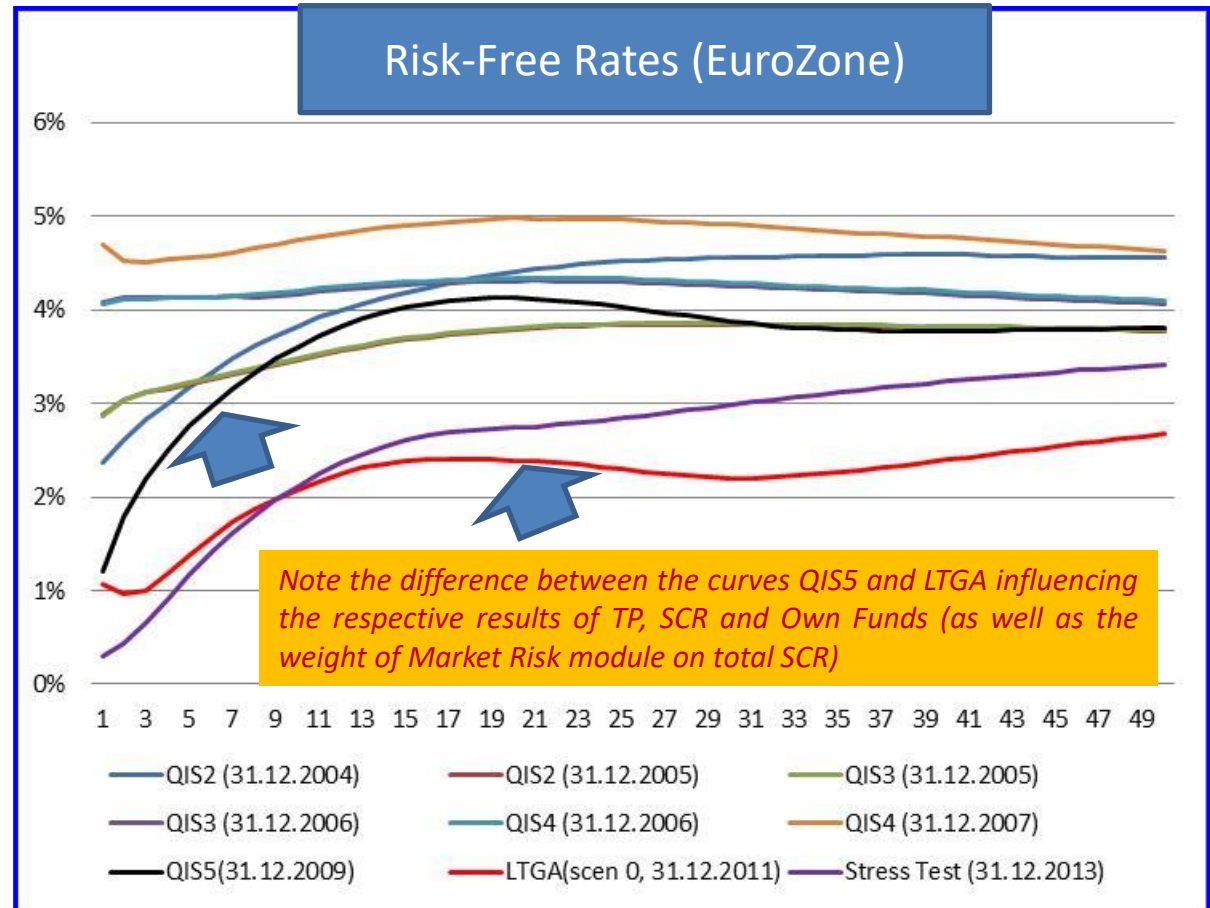


3.2. Discounting Rates

Discounting Rates

Risk-Free Rates

- For a given currency and valuation date, each insurance and reinsurance undertaking **should use the same relevant basic risk-free interest rate term structures**
- Basic risk-free rates provided by QIS and stress tests are reported in Figure (some QIS provided the curve at the relevant date and one year before)
- Curves here reported do not include volatility adjustments (described later)



State of the art

(DA and Technical documents)

- After the publication of Delegated Acts, **EIOPA is doing some refinements (last updates September 2021)** to the calculation of interest rate and volatility adjustment.
- According to the *Technical documentation of the methodology to derive EIOPA's risk-free interest rate term structures* and to Commission Implementing Regulation 2016/165, we have:
 - **A basic free interest rate curve** calculated for different maturities and for different currencies
 - calibrated mainly by swap (where available),
 - corrected to remove credit risk,
 - based on data until a last liquid point (fixed by EIOPA) and then extrapolated towards an ultimate forward rate (fixed by EIOPA)
 - An adjustment (**volatility adjustment**) to the liquid part of the risk-free rate in order to reduce the impact of short-term market volatility
 - provided by EIOPA
 - currency and country based
 - A different adjustment (**the matching adjustment**) is made in respect of predictable portfolios of liabilities.
 - An undertaking can assign to eligible portfolios assets with fixed cash flows that it intends to hold to maturity.
 - EIOPA is required to provide an estimate of what portion of the spread of such assets above the risk-free interest rate reflects risks not faced by those who hold assets to maturity.

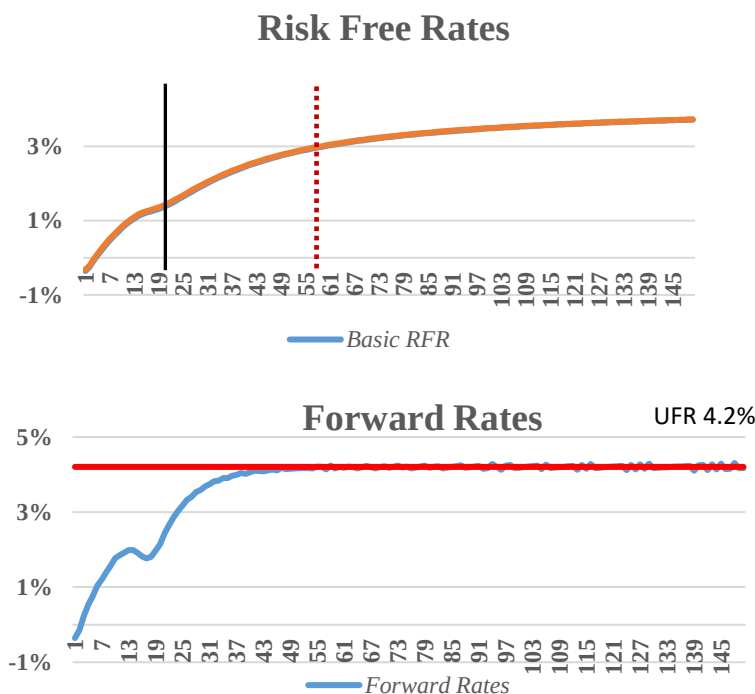
Risk-Free rates curve at 31.12.2017

Risk-Free Curve is based on following step

- **calibrated mainly by swap** (where available, if not use Govies) **until a last liquid point (LLP)** (20 years for Euro curve)
- **Adjustment to swap rates for credit risk (CRA)**. The credit risk adjustment shall be determined on the basis of the difference between swap rates and Overnight Index Swap (parallel downward shift applied until LLP). equal to 50% of the average of that difference over a time period of one year. The adjustment shall not be lower than 10 basis points and not higher than 35 basis points.
- **Smith-Wilson extrapolation method for estimation of rates of longer maturities** (greater than LLP). **Convergence period and Ultimate Forward Rate (UFR) are given.** UFR equal to 4.2% until the end of 2017 (4.05% since January 2018, 3.90% since January 2019, 3.75% since January 2020, 3.60% since January 2021, 3.45% for 2022 and 2023).

Fonte : Technical documentation of the methodology to derive EIOPA's risk-free interest rate term structures, 2018

Spot and Forward Rates (Euro) at the end of 2017.



Volatility Adjustment (VA)

- **Volatility Adjustment (VA)** is designed to protect insurers with long-term liabilities from the impact of **volatility** on the insurers' solvency position
- **It is a constant addition to the basic risk-free curve.**
It is added to the liquid part of the risk-free zero-coupon rates, i.e. until the so-called Last Liquid Point (LLP). After the LLP, the curve converges to the UFR. The resulting rates are used to produce the relevant risk-free curve. The VA is provided and updated by EIOPA and can differ for each major currency and country.
- It's based on 65% of a risk corrected spread ($S_{currency}^{RC}$)
- Risk corrected spread is based on the difference between:
 - **The interest rate that could be earned from a reference portfolio** (representative of the investments made by insurance and reinsurance undertakings in that currency to cover the best estimate for insurance and reinsurance obligations denominated in that currency)
 - **The basic risk-free interest rates**

It's corrected to consider the portion of the currency spread S , denoted RC for risk correction, which corresponds to "the portion of the spread that is attributable to a realistic assessment of expected losses, unexpected credit risk or any other risk, of the assets" in the reference portfolio ($S_{currency}^{RC} = S - RC$)

$$VA_{total} = \underbrace{0,65 * [S_{currency}^{RC}]}_{\text{Risk corrected spread (currency-specific)}} + \underbrace{(S_{country}^{RC} > 1\%) * \max(S_{country}^{RC} - 2 * S_{currency}^{RC}; 0)}_{\text{Risk corrected spread (country-specific)}}$$

31.12.2017: Volatility Adjustment: 4 bp until 22nd year and then decreasing to zero

31.12.2018, VA: 24 bp, 31.12.2019, VA: 7 bp, 31.12.2020 VA: 7 bp, 31.12.2021 VA: 3 bp, 30.06.2022 VA 25 bp

Matching Adjustment

- The Matching Adjustment (MA) is a parallel shift applied to the entire basic risk-free term structure and serves the same purpose as the VA. The MA is calculated on the basis of the match between the insurers' assets and the liabilities (must be approved by the Supervisor). The MA is corrected for the fundamental spread (given by EIOPA).
- **The MA can only be applied to a portfolio of life insurance obligations with an assigned portfolio of assets that covers the best estimate of the liabilities.** The mismatch between the cash flows of the assets and the cash flows of the liabilities must not be a material risk in relation to the risks inherent to the insurance business. **These portfolios need to be identified, organized and managed separately from other activities of the insurers.** Furthermore, the assigned portfolio of assets cannot be used to cover losses arising from other activities of the insurers.
- Insurance portfolio must satisfy some constraints (Art. 77b Directive 2014/51):
 - the only underwriting risks connected to the portfolio of insurance or reinsurance obligations are longevity risk, expense risk, revision risk and mortality risk;
 - where the underwriting risk connected to the portfolio of insurance or reinsurance obligations includes mortality risk, the best estimate of the portfolio of insurance or reinsurance obligations does not increase by more than 5 % under a mortality risk stress that is calibrated in accordance with Article 101(2) to (5);
 - the contracts underlying the portfolio of insurance or reinsurance obligations do not give rise to future premium payments;
 - the contracts underlying the portfolio of insurance or reinsurance obligations include no options for the policy holder or only a surrender option where the surrender value does not exceed the value of the assets, valued in accordance with Article 75, covering the insurance or reinsurance obligations at the time the surrender option is exercised;
- *MA is based on the difference of (art. 77c, Direttiva 2014/51):*
 - **the annual effective rate**, calculated as the single discount rate that, where applied to the cash flows of the portfolio of insurance or reinsurance obligations, results in a value that is equal to the value **of the portfolio of assigned assets**;
 - **the annual effective rate**, calculated as the single discount rate that, where applied to the cash flows of the portfolio of insurance or reinsurance obligations, **results in a value that is equal to the value of the best estimate of the portfolio of insurance or reinsurance obligations** where the time value of money is taken into account using the basic risk-free interest rate term structure;
- **the matching adjustment must not include the fundamental spread reflecting the risks retained by the insurance or reinsurance undertaking;**

An Example

Not Discounted Cash-Flow

Sifa Insurer (Paid CHL method)

	Ultimate Cost	Latest Cumulative Payments	Undiscounted Claims Reserve		Expected Future Cash-Flow (Claims Payments)
1993	86,842.701	85,775	1,068.088	2005	105,554.571
1994	98,358.305	96,325	2,033.283	2006	47,325.382
1995	107,662.200	105,007	2,654.715	2007	25,965.122
1996	118,351.459	114,674	3,677.046	2008	17,130.394
1997	123,149.426	118,289	4,859.965	2009	12,745.385
1998	139,128.238	132,139	6,989.667	2010	9,765.250
1999	147,730.791	138,291	9,439.777	2011	7,737.803
2000	141,900.767	129,900	12,000.357	2012	6,005.343
2001	140,813.425	124,786	16,027.344	2013	4,720.533
2002	146,781.262	121,800	24,981.508	2014	3,847.964
2003	160,933.257	112,342	48,591.257	2015	3,439.780
2004	174,421.018	60,361	114,059.742	2016	2,145.224
			246,382.750		246,382.750

BE of claims reserve

$$BE = \sum_{i=0}^n \sum_{j=n-i+1}^N \frac{\hat{P}_{i,j}}{(1 + r(0, j - n + i))^{j-n+i}} = \sum_{t=1}^N \frac{E(CF_{n+t})}{(1 + r(0, t))^t}$$

N : last development year ($N = n$ if no tail)
 $E(CF_{n+t}) = \sum_{i=t}^n \hat{P}_{i,n-i+t}$ (if no tail is present)

Calendar Year	Discounted Cash-Flow							
	QIS2	QIS3	QIS4	QIS5	LTGA	end 2016	end 2017	june 2018
2005	102,605.68	101,421.64	100,820.06	104,020.11	103,419.27	105,736.44	105,891.31	105,822.30
2006	44,585.33	43,652.25	43,315.51	45,441.91	45,521.59	47,449.62	47,524.78	47,491.46
2007	23,680.97	23,001.32	22,746.79	24,140.59	24,463.12	26,025.97	26,002.55	25,993.18
2008	15,124.17	14,574.34	14,346.75	15,356.46	15,708.90	17,125.60	17,055.91	17,051.14
2009	10,882.89	10,413.50	10,201.69	10,982.68	11,329.91	12,678.05	12,587.88	12,577.84
2010	8,055.00	7,659.86	7,464.48	8,067.16	8,385.00	9,636.18	9,541.54	9,529.57
2011	6,160.57	5,823.77	5,642.85	6,113.81	6,404.72	7,553.49	7,467.63	7,449.46
2012	4,609.82	4,344.78	4,173.78	4,529.10	4,789.52	5,783.78	5,713.35	5,692.50
2013	3,488.40	3,274.15	3,122.94	3,392.62	3,624.89	4,476.71	4,419.44	4,397.79
2014	2,737.40	2,557.34	2,421.22	2,632.18	2,842.54	3,588.34	3,538.48	3,522.03
2015	2,351.81	2,189.51	2,057.19	2,237.74	2,441.24	3,150.77	3,104.39	3,087.52
2016	1,409.51	1,307.61	1,219.43	1,326.75	1,461.73	1,928.84	1,898.81	1,885.55
BE	225,691.552	220,220.076	217,532.682	228,241.112	230,392.419	245,133.787	244,746.063	244,500.339
BE/Ultimate Cost	91.60%	89.38%	88.29%	92.64%	93.51%	99.49%	99.34%	99.24%

Hp: Payments discounted at the end of the year

QIS2-QIS4: $r(0,t)=rf(0,t)$

QIS5: $r(0,t)=rf(0,t)+50\%lp$

LTGA= $r(0,t)=rf(0,t)+CCP$ (CCP=100bp)

BE2016-18 $r(0,t)=rf(0,t)+va$

Duration of Liabilities

- As for a financial asset (like bonds for instance), it is possible to compute the duration or the convexity of claims reserve.

- For instance, Duration of a single accident year i , is equal to:

$$D_i = \frac{\sum_{j=n-i+1}^N (j - n + i) \hat{P}_{i,j} (1 + r(0, j - n + i))^{-(j-n+i)}}{\sum_{j=n-i+1}^N \hat{P}_{i,j} (1 + r(0, j - n + i))^{-(j-n+i)}}$$

- It is worth noting that the denominator is the Best Estimate of accident year i
- Obviously, also modified duration can be easily derived. It gives the percentage change in value of claims reserve for a given change in interest rates (parallel shift).

An example

Accident year	Duration
1,993	1.00
1,994	2.19
1,995	2.34
1,996	2.85
1,997	3.24
1,998	3.53
1,999	3.76
2,000	3.82
2,001	3.81
2,002	3.52
2,003	2.95
2,004	2.34
Total	2.85

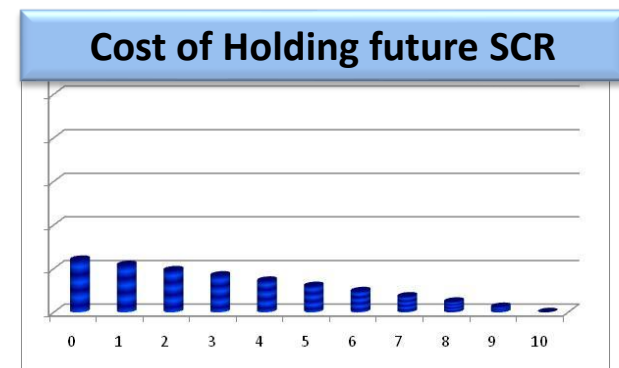
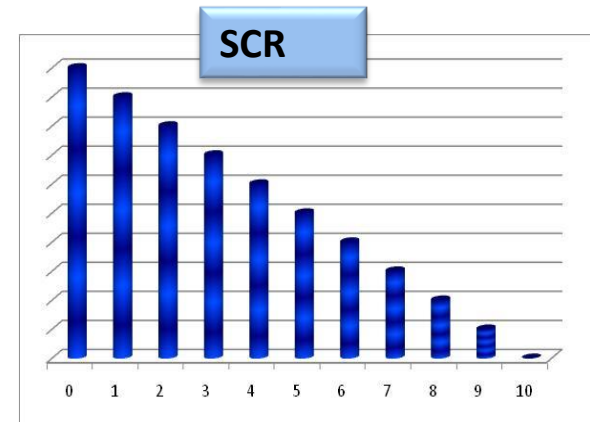
Calendar Year	Discounted CF	Weighted CF	
2005	105,891.31	105,891.31	
2006	47,524.78	95,049.55	
2007	26,002.55	78,007.64	
2008	17,055.91	68,223.63	
2009	12,587.88	62,939.42	
2010	9,541.54	57,249.25	
2011	7,467.63	52,273.44	
2012	5,713.35	45,706.78	
2013	4,419.44	39,774.99	
2014	3,538.48	35,384.78	
2015	3,104.39	34,148.29	
2016	1,898.81	22,785.69	Duration
Total	244,746.06	697,434.77	2.85

Example has been computed using discounting rates at the end of 2017

3.3. Risk Margin

Risk Margin

- The **risk margin** is obtained by:
 - projecting the SCR** until the complete run-off of the overall liabilities (*diversification assumed*).
 The SCR captures only some risks:
 - **underwriting risk** (only existing business);
 - **default risk** with respect to reinsurance contracts and any other material exposures which are closely related to the insurance and reinsurance obligations;
 - **operational risk**;
 - **material market risk** (unavoidable market risk in QIS5).
 - quantifying **the cost of capital of each year** ($6\% \cdot SCR_t$);
 - discounting** using the basic risk-free rates (without corrections for va or ma).



QIS5 results show (for Solo Insurers) a ratio RM/BE for Non Life LoBs equal to:

	Italy	EIOPA
MTPL	4%-5%	5.62%
GTPL	4%-5%	6.83%
MOD	6%-7%	7.85%

$$RM = \sum_{t=0}^T CoC \cdot v(0, t+1) \cdot SCR_t$$

Proxy and simplification for Risk Margin

- ❑ The following simplified methods can be used to derive **Risk Margin**:
 - **Approximate the individual risks or sub-risks** within some or all modules and sub-modules to be used for the calculation of future SCRs;
 - **Approximate the whole SCR** for each future year, e.g. **by using a proportional approach**;
 - Estimate all future SCRs “at once”, e.g. **by using an approximation based on the duration approach**;
 - Approximate the risk margin **by calculating it as a percentage of the best estimate**.

- ❑ QIS5 report shows that:
 - **40% of Non-Life insurers** used a proportional approach;
 - **20 % of Non-Life insurers** used a percentage of best estimate.

Simplification of Risk Margin:

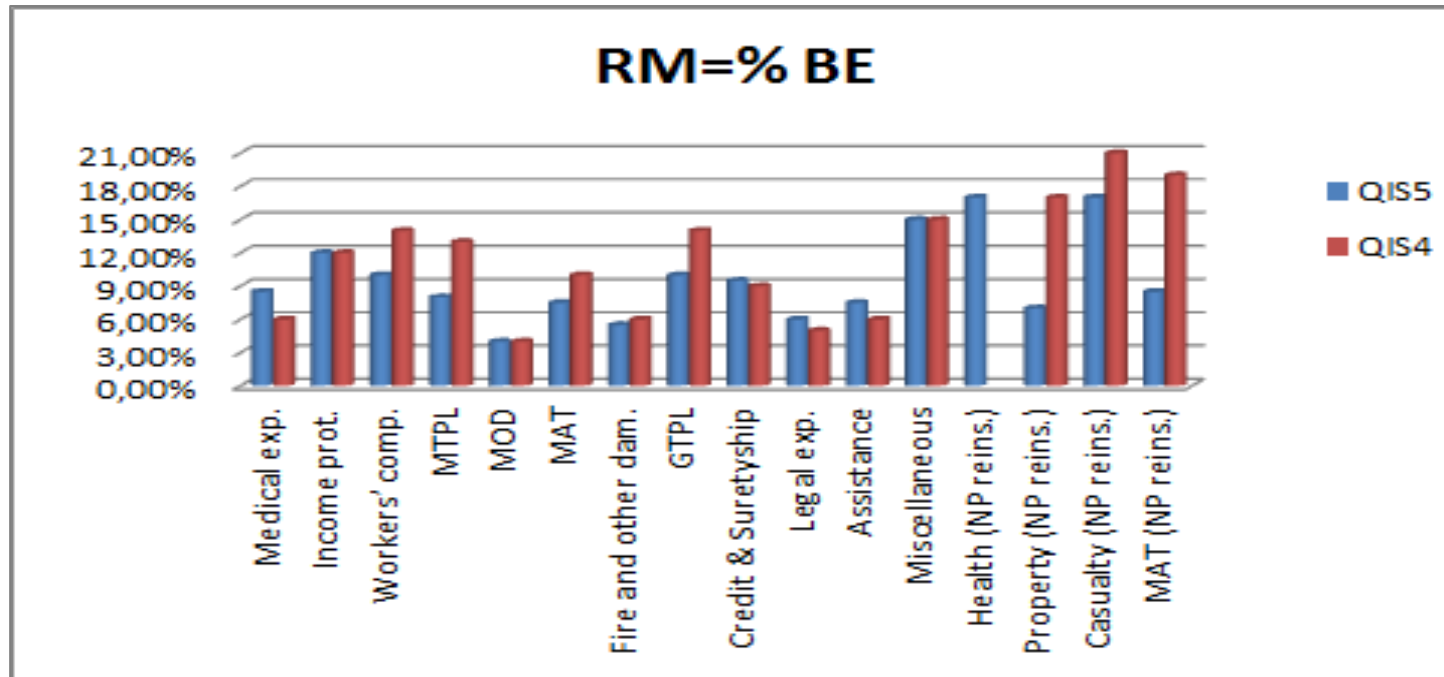
Proportional rule

- ❑ This simplification is based on the assumption that the future SCR_t are proportional to the best estimate of technical provisions for the relevant year.
- ❑ The proportionality factor, used for the estimation of SCR_t ($t > 0$), is the ratio of net (of reinsurance) best estimate at time t to net best estimate at time 0.

$$SCR_t = SCR_0 \cdot \frac{BE_t^{net}}{BE_0^{net}}$$

Risk Margin Proxy

Percentage Approach



❑ It is noteworthy that:

- Medical Expenses and Income Protection had a different granularity in QIS4 (Health (6%) and Accident (12%));
- NP reinsurance Health has been introduced in QIS5.

Risk Margin/T.P. Solo Entities

Italian market (96 Insurers in 2018)

	2016	2017	2018
Non-Life	4.46%	3.96%	3.89%
Health (non SLT)	4.47%	4.48%	4.29%
Health (SLT)	13.0%	7.15%	10.7%
Life (not linked)	0.72%	0.79%	0.97%

EEA (2790 Insurers in 2018)

	2016	2017	2018
Non-Life	5.69%	5.36%	5.25%
Health (non SLT)	6.28%	6.52%	6.61%
Health (SLT)	5.43%	5.45%	5.78%
Life (not linked)	1.64%	1.56%	1.67%